Improved model reduction with basis enrichment for dynamic analysis of nearly periodic structures including substructures with geometric changes

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Abstract

Model reduction based on matrix interpolation provides an efficient way to compute the dynamic response of nearly periodic structures composed of substructures (cells) with varying properties. This may concern 2D or 3D substructures subjected to geometric modifications (mesh variations) or more classic dimension changes. An efficient interpolation strategy for a nearly periodic structure can be obtained by (i) interpolating reduced substructure matrices over a multi-dimensional parametric space and (ii) reducing the number of degrees of freedom at substructure boundaries by considering the interface modes of an equivalent purely periodic structure. In this paper, two basis enrichment techniques are proposed to improve the accuracy of the interpolation strategy. This consists in (i) considering high-order static modes, in addition to component modes, to express the reduced substructure matrices and (ii) adding static correction vectors into the basis of interface modes to account for the varying properties of the substructures. Numerical experiments are carried out which clearly highlight the relevance of the basis enrichment techniques for predicting the harmonic behavior of nearly periodic structures with 2D or 3D substructures.

Key words: nearly periodic structures, geometric changes, reduced models, matrix interpolation, basis enrichment.

1. Introduction

Periodic structures are frequently encountered in mechanical engineering, e.g., in the aeronautic (fuselages, turbines) and railway industries. It is well known that these types of structures possess band gap properties, i.e., the fact that they do not convey energies on some frequency regions (band gaps) which as such appear interesting for the passive control of vibrations. These structures are usually called metamaterials, representing for instance structures with 2D-periodicity composed of resonant substructures where band gaps occur in the vicinity of their resonance frequencies. There exist plenty of works in the literature about wave propagation (e.g., Bloch waves) in 2D periodic structures and the related band gap effect [1–8]. However, the development of numerical models capable of quickly predicting the forced response of these structures is much less reported. Especially the development of wave-based models for 2D periodic structures of finite dimensions is prone to tough issues which make this task difficult, even impossible to carry out. These issues are mostly linked to the multi-directional behavior of the propagating waves and the description of the wave reflection phenomena at boundaries that arise from these multi-directional waves. Component mode synthesis (CMS) techniques represent a good alternative for modeling these structures and providing reduced order models which can be efficiently built and simulated. CMS includes the Craig-Bampton (CB) and the Rubin or MacNeal methods where

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the displacement vectors of the substructures are described using either static and fixed interface modes (CB), or attachment and free interface modes (Rubin, MacNeal) [9]. Within the CMS framework, reduced mass, damping and stiffness matrices of substructures are obtained via Galerkin projection of the finite element (FE) matrices onto a set of component modes (static and fixed interface modes, or attachment and free interface modes). Note that, since the substructures are identical for periodic structures, their component modes and reduced matrices only need to be computed once. To further improve the efficiency of CMS techniques, a reduction of the number of degrees of freedom (DOFs) at the interface between the substructures can be invoked, see for instance [10–12].

Aside from purely periodic structures, nearly periodic structures involve disordered substructures with varying properties. These disordered substructures can result from geometric variability due to manufacturing process, or they can be deliberately introduced for the design of structures with particular properties. Indeed nearly periodic structures are likely to provide additional benefits like energy localization effects [13–16] which could help improve the vibration control of systems. In [17], a general framework to model 2D continuous substructures with geometric changes has been proposed. This consists in considering distorted substructure FE meshes by moving the nodes of a baseline mesh via shape functions and mesh parameters. The mesh parameters are different between the substructures and, therefore, they can be chosen in a random way to generate different substructures. To fix ideas, two nearly periodic structures with 2D and 3D substructures are shown in Fig. 1.

To quickly recompute the reduced substructure matrices, for any new parametric variation, an interpolation strategy can be used. Model reduction by matrix interpolation for systems with parametric dependency is a research topic which is well documented in the literature, see [18–21]. To interpolate reduced matrices over a parametric space, they need to be expressed at interpolation points in compatible coordinate systems [18]. In [17], this strategy has been adapted to substructures with geometric changes, where the reduced substructure matrices are formulated using the CB method. To express the reduced matrices in compatible coordinate systems, a modal assurance criterion (MAC) [22] for correlating the reduced matrices of fixed interface modes between different interpolation points has been proposed. Also in [17], an interface reduction (between the substructures) for modeling a nearly periodic structure has been considered in the same way as in [10–12] but where the projection basis consists of the interface modes of the equivalent purely periodic structure.

This strategy based on the interpolation of the reduced substructure matrices is prone to numerical limitations and lack of accuracy, however. This first concerns the limited number of fixed interface modes which are retained in the CB basis of the substructures. Indeed, although it seems straightforward to correlate CB bases with low-order (regular) fixed interface modes between different interpolation points, the procedure is likely to fail when high-order modes are added. Next the proposed interface reduction for modeling a nearly periodic structure is based on the interface modes of a purely periodic one, which as such are not able to account for geometric modifications. To solve these issues and improve the accuracy of the interpolation strategy, two basis enrichment techniques are proposed in this paper whose key steps can be detailed as follows:

- Enrichment of the CB basis of the substructures using high-order static modes. The idea stems from the works in [23, 24], and it is applied here to express interpolated reduced substructure matrices. Prior to any matrix interpolation, these modes are orthogonalized and transformed using a MAC-based correlation criterion (as reported earlier). Then a reduced set of these high-order static modes are used to enrich the CB basis at the interpolation points.

- Enrichment of the basis of interface modes using static correction vectors. The strategy originates from the works in [25, 26]. It consists in adding to the interface modes a set of static response vectors resulting from parametric changes, i.e., those induced by the perturbations of the stiffness matrices of the substructures.

From the author's point of view, these basis enrichment techniques are fairly new within the topics of substructuring and matrix interpolation. These techniques can make possible the dynamic responses of nearly periodic structures with complex substructures to be predicted with accurate precision at low and high frequencies, see Sec. 4.

The rest of the paper is organized as follows. In Sec. 2, a background about the modeling of nearly periodic structures using interpolated reduced substructure matrices is provided. Substructure models depending on an arbitrary number of parameters (mesh distortions, ...) are considered. Also, several generalizations and improvements of the approach proposed in [17] are brought. These concern (i) the correlation criterion for expressing the reduced matrices of fixed interface modes of the substructures (Sec. 2.1) and (ii) the use of residual flexibility vectors to enrich the basis of interface modes. In Sec. 3, the basis enrichment techniques are detailed. The strategy that uses high-order static modes to enrich the CB basis of the substructures is detailed in Sec. 3.1; also the use of static correction vectors to improve the accuracy of the basis of interface modes is discussed in Sec. 3.2. In Sec. 4, numerical experiments are carried out where the advantages of the interpolation strategy with basis enrichment over the classic interpolation strategy are clearly highlighted. Three examples featuring nearly periodic structures with different types of substructures are analyzed. Secs. 4.1 and 4.2 focus on 2D substructures whose FE models depend on two and three parameters, respectively; Sec. 4.3 focuses on 3D substructures with FE models of relative large size depending on three mesh parameters.

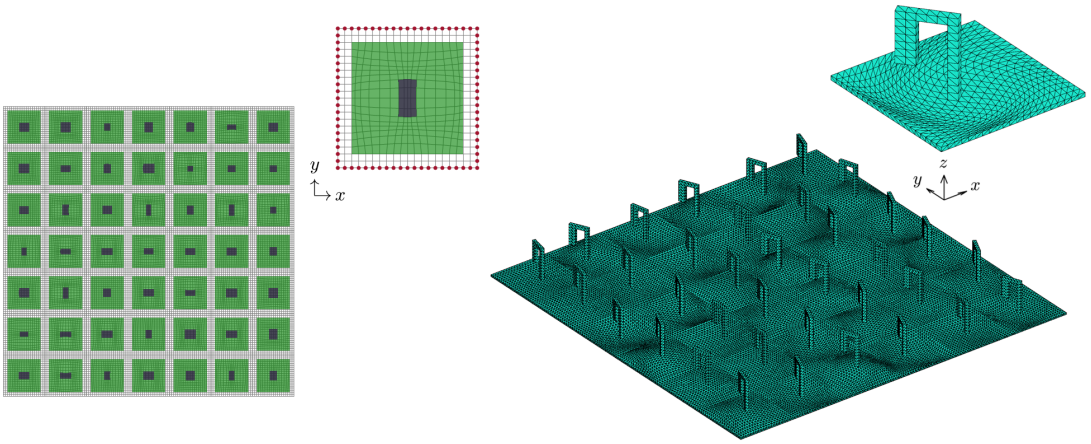


Figure 1: Examples of nearly periodic structures with 2D and 3D substructures.

2. Model reduction based on matrix interpolation: background

2.1. Reduced modeling of substructures

Examples of nearly periodic structures with 2D and 3D substructures undergoing geometric changes are shown in Fig. 1. The focus here is on geometric variations in the $(x, y)-$plane which are introduced by considering distorted FE meshes [17]. For a substructure s, a distorted mesh is generated by moving the positions of the nodes of a baseline/undistorted mesh (x_j^0, y_j^0) as follows:

$$x_j^s = x_j^0 + \sum_{k=1}^{n^x} \epsilon_{xk}^s f_{xk}(x_j^0, y_j^0) \quad , \quad y_j^s = y_j^0 + \sum_{k=1}^{n^y} \epsilon_{yk}^s f_{yk}(x_j^0, y_j^0), \tag{1}$$

where $f_{xk}(x,y)$ and $f_{yk}(x,y)$ are shape functions — identical for all the substructures — which vanish at the boundary of the substructure [1]. Also, in Eq. (1), ϵ^s_{xk} $(k = 1,\ldots,n^x)$ and ϵ^s_{yk} $(k = 1,\ldots,n^y)$ are real scalars referred to as mesh parameters. These parameters are assumed to be different between the substructures and can be defined in a completely arbitrary way, e.g., to generate structures with tunable properties [27]. In the present study, ϵ^s_{xk} and ϵ^s_{yk} are supposed to represent independent variables defined on supports $[-\delta_{xk}\,,\,\delta_{xk}]$ and $[-\delta_{yk}\,,\,\delta_{yk}]$ with dispersion parameters δ_{xk} and δ_{yk}. The mesh transformation provided by Eq. (1) covers a large variety of problems for substructures subjected to various kinds of geometric changes as shown for instance in Fig. 2 for some abstract 2D substructure with distorted FE meshes.

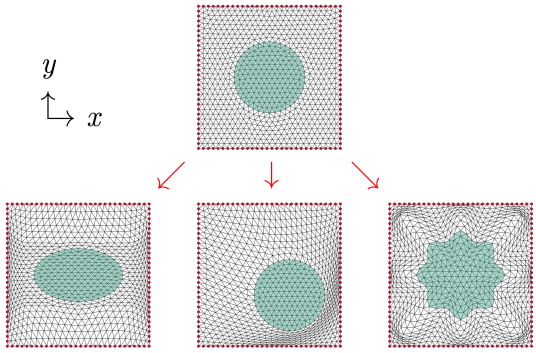


Figure 2: 2D substructure, including a circular core (in green), with various geometric changes (red dots represent boundary nodes).

Eq. (1) introduces several shape functions and mesh parameters $(f_{xk}(x,y),\epsilon^s_{xk})$ $(k = 1,\ldots,n^x)$ and $(f_{yk}(x,y),\epsilon^s_{yk})$ $(k = 1,\ldots,n^y)$ for describing a distorted substructure. This constitutes a generalization of the strategy proposed in [17] where only two shape functions and two mesh parameters (per substructure) were considered. Besides the substructure models may also depend on classic parameters θ^s_k $(k = 1,\ldots,n^\theta)$ that could represent any constant dimensions (e.g., thicknesses) [2], defined as:

$$\theta^s_k = \theta^0_k + \epsilon^s_{\theta k} \qquad k = 1,\ldots,n^\theta, \tag{2}$$

for some nominal values θ^0_k and independent variables $\epsilon^s_{\theta k}$ defined on supports $[-\delta_{\theta k}\,,\,\delta_{\theta k}]$. Thus the FE-based dynamic equation of a substructure s subjected to geometric changes can be written in the frequency domain (frequency ω) as:

$$\left(-\omega^2\mathbf{M}(\boldsymbol{\epsilon}^s) + \mathtt{i}\omega\mathbf{C}(\boldsymbol{\epsilon}^s) + \mathbf{K}(\boldsymbol{\epsilon}^s)\right)\mathbf{u}^s = \mathbf{F}^s, \tag{3}$$

where $\mathbf{u}^s$ and $\mathbf{F}^s$ are displacement and force vectors (respectively), and $\mathbf{M}(\boldsymbol{\epsilon}^s)$, $\mathbf{C}(\boldsymbol{\epsilon}^s)$ and $\mathbf{K}(\boldsymbol{\epsilon}^s)$ are mass, damping and stiffness matrices (respectively). Here, $\boldsymbol{\epsilon}^s$ represents the vector of parameters defined as $\boldsymbol{\epsilon}^s = [(\boldsymbol{\epsilon}^s_x)^T(\boldsymbol{\epsilon}^s_y)^T(\boldsymbol{\epsilon}^s_\theta)^T]^T$ with $\boldsymbol{\epsilon}^s_x = [\epsilon^s_{x1}\cdots\epsilon^s_{xn^x}]^T$ and $\boldsymbol{\epsilon}^s_y = [\epsilon^s_{y1}\cdots\epsilon^s_{yn^y}]^T$ the vector of mesh parameters (distortion), and $\boldsymbol{\epsilon}^s_\theta = [\epsilon^s_{\theta 1}\cdots\epsilon^s_{\theta n^\theta}]^T$ the vector of parameters that concern other parametric changes. In Eq. (3), $\mathbf{C}(\boldsymbol{\epsilon}^s)$ is the damping matrix which for instance can be of Rayleigh type, i.e.,

$$\mathbf{C}(\boldsymbol{\epsilon}^s) = a\mathbf{M}(\boldsymbol{\epsilon}^s) + b\mathbf{K}(\boldsymbol{\epsilon}^s), \tag{4}$$

[1] The boundary nodes are highlighted by red dots for the left substructure in Fig. 1 (see also Fig. 2).

[2] Note that the procedure can be generalized without difficulty to substructures with varying material properties, e.g., density and Young's modulus.

with a and b two positive real constants which are similar for all the substructures.

Reduced models of the substructures can be expressed using the CB method [28, 29]. The classic CB approximation for substructures with parameter-independent matrices $\mathbf{M}$ and $\mathbf{K}$ (and $\mathbf{C}$) writes:

$$\mathbf{u}^s = \begin{bmatrix} \mathbf{u}^s_{\mathtt{B}} \\ \mathbf{u}^s_{\mathtt{I}} \end{bmatrix} \approx \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{X}_{\mathtt{st}} & \widetilde{\mathbf{X}} \end{bmatrix} \begin{bmatrix} \widetilde{\mathbf{u}}^s_{\mathtt{B}} \\ \widetilde{\boldsymbol{\alpha}}^s \end{bmatrix} = \widetilde{\mathbf{T}} \widetilde{\mathbf{u}}^s, \tag{5}$$

where subscripts $\mathtt{B}$ and $\mathtt{I}$ refer to boundary and internal DOFs, respectively. In Eq. (5), $\widetilde{\mathbf{T}}$ is the CB transformation matrix, $\mathbf{X}_{\mathtt{st}}$ is the matrix of static modes defined by $\mathbf{X}_{\mathtt{st}} = -\mathbf{K}_{\mathtt{II}}^{-1}\mathbf{K}_{\mathtt{IB}}$, and $\widetilde{\mathbf{X}}$ is the reduced matrix of fixed interface modes — i.e., the eigenvectors of the matrix pencil $(\mathbf{K}_{\mathtt{II}}, \mathbf{M}_{\mathtt{II}})$ — where only a small number $M_{\mathtt{I}}$ of these modes are retained, see [17]; also, in Eq. (5), $\widetilde{\boldsymbol{\alpha}}^s$ represents the $M_{\mathtt{I}} \times 1$ vector of generalized coordinates for the fixed interface modes. The reduced dynamic equations of the substructures follow from projecting the matrices $\mathbf{M}$, $\mathbf{K}$ and $\mathbf{C}$, and the vector $\mathbf{F}$, onto the column space of $\widetilde{\mathbf{T}}$, i.e.,

$$\left(-\omega^2 \widetilde{\mathbf{M}} + \mathtt{i}\omega \widetilde{\mathbf{C}} + \widetilde{\mathbf{K}}\right) \widetilde{\mathbf{u}}^s = \widetilde{\mathbf{F}}^s, \tag{6}$$

where

$$\widetilde{\mathbf{M}} = \widetilde{\mathbf{T}}^T \mathbf{M} \widetilde{\mathbf{T}} \quad , \quad \widetilde{\mathbf{K}} = \widetilde{\mathbf{T}}^T \mathbf{K} \widetilde{\mathbf{T}} \quad , \quad \widetilde{\mathbf{C}} = a\widetilde{\mathbf{M}} + b\widetilde{\mathbf{K}} \quad , \quad \widetilde{\mathbf{F}}^s = \widetilde{\mathbf{T}}^T \mathbf{F}^s. \tag{7}$$

For the sake of simplicity, it will be assumed here and in the rest of the paper that the internal DOFs of the substructures are free from excitation. This gives:

$$\mathbf{F}^s = \begin{bmatrix} \mathbf{F}^s_{\mathtt{B}} \\ \mathbf{F}^s_{\mathtt{I}} \end{bmatrix} = \begin{bmatrix} \mathbf{F}^s_{\mathtt{B}} \\ \mathbf{0} \end{bmatrix} \qquad \text{and} \qquad \widetilde{\mathbf{F}}^s = \widetilde{\mathbf{T}}^T \mathbf{F}^s = \begin{bmatrix} \mathbf{I} & \mathbf{X}^T_{\mathtt{st}} \\ \mathbf{0} & \widetilde{\mathbf{X}}^T \end{bmatrix} \begin{bmatrix} \mathbf{F}^s_{\mathtt{B}} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{F}^s_{\mathtt{B}} \\ \mathbf{0} \end{bmatrix}. \tag{8}$$

Concerning now substructures with parameter-dependent matrices $\mathbf{M}(\boldsymbol{\epsilon}^s)$ and $\mathbf{K}(\boldsymbol{\epsilon}^s)$ (and $\mathbf{C}(\boldsymbol{\epsilon}^s)$), the CB method raises the issue of computing the transformation matrices $\widetilde{\mathbf{T}} = \widetilde{\mathbf{T}}(\boldsymbol{\epsilon}^s)$ and the reduced matrices $\widetilde{\mathbf{M}} = \widetilde{\mathbf{M}}(\boldsymbol{\epsilon}^s)$ and $\widetilde{\mathbf{K}} = \widetilde{\mathbf{K}}(\boldsymbol{\epsilon}^s)$ via Eq. (7) several times. This can quickly turn computationally cumbersome as soon as the number of substructures and the number of parameters $\boldsymbol{\epsilon}^s$ analyzed are becoming large. To solve this issue, the interpolation strategy proposed in [17] can be considered. For the sake of clarity, the main steps of the strategy are reviewed hereafter.

To avoid computing the reduced matrices many times, they are only expressed for a few sets of parameters $\boldsymbol{\epsilon}^s = \boldsymbol{\epsilon}_p$ with $p = 1, \ldots, n^p$ and n^p small. Thus the reduced substructure matrices for any set of parameters $\boldsymbol{\epsilon}^s$ can be approximated between the interpolation points "$\boldsymbol{\epsilon}_p$" by matrix interpolation, where each vector $\boldsymbol{\epsilon}_p$ concerns particular parameters $\boldsymbol{\epsilon}^s_x = (\boldsymbol{\epsilon}_x)_p$, $\boldsymbol{\epsilon}^s_y = (\boldsymbol{\epsilon}_y)_p$ and $\boldsymbol{\epsilon}^s_\theta = (\boldsymbol{\epsilon}_\theta)_p$. At these interpolation points, the transformation matrix and the reduced matrices are expressed as:

$$\widetilde{\mathbf{T}}_p = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ (\mathbf{X}_{\mathtt{st}})_p & \widetilde{\mathbf{X}}_p \end{bmatrix} \quad , \quad \widetilde{\mathbf{M}}_p = \widetilde{\mathbf{T}}_p^T \mathbf{M}(\boldsymbol{\epsilon}_p) \widetilde{\mathbf{T}}_p \quad , \quad \widetilde{\mathbf{K}}_p = \widetilde{\mathbf{T}}_p^T \mathbf{K}(\boldsymbol{\epsilon}_p) \widetilde{\mathbf{T}}_p, \tag{9}$$

with $(\mathbf{X}_{\mathtt{st}})_p = -\mathbf{K}_{\mathtt{II}}(\boldsymbol{\epsilon}_p)^{-1}\mathbf{K}_{\mathtt{IB}}(\boldsymbol{\epsilon}_p)$ the matrix of static modes at the interpolation points, and $\widetilde{\mathbf{X}}_p$ the matrix of fixed interface modes representing a reduced set of eigenvectors of the matrix pencil $(\mathbf{K}_{\mathtt{II}}(\boldsymbol{\epsilon}_p), \mathbf{M}_{\mathtt{II}}(\boldsymbol{\epsilon}_p))$. These modes can be classically normalized in such a way that $\widetilde{\mathbf{X}}_p^T \mathbf{M}_{\mathtt{II}}(\boldsymbol{\epsilon}_p) \widetilde{\mathbf{X}}_p = \mathbf{I}$. A straightforward selection of the interpolation points along each parametric space $[-\delta_{xk}\ ,\ \delta_{xk}]$, $[-\delta_{yk}\ ,\ \delta_{yk}]$ or $[-\delta_{\theta k}\ ,\ \delta_{\theta k}]$ is to use Gauss points (see discussion below). As for the interpolation functions, they can be simply defined as Lagrangian polynomials. Although easy to understand, the procedure for interpolating matrices however requires the reduced matrices $\widetilde{\mathbf{M}}_p$ and $\widetilde{\mathbf{K}}_p$ to be expressed in compatible

coordinate systems, see [18]. For this task, the following alternative matrices of fixed interface modes, at the interpolation points, can be considered:

$$\widehat{\mathbf{X}}_p = \widetilde{\mathbf{X}}_p(\boldsymbol{\Psi}^T\widetilde{\mathbf{X}}_p)^{-1}, \tag{10}$$

for some well-chosen matrix $\boldsymbol{\Psi}$. In [17], it was proposed to define $\boldsymbol{\Psi}$ as $\boldsymbol{\Psi} = (\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\widetilde{\mathbf{X}}^0$ where notations $\mathbf{M}^0_{\mathtt{II}}$ and $\widetilde{\mathbf{X}}^0$ refer to $\mathbf{M}^s_{\mathtt{II}}$ and $\widetilde{\mathbf{X}}^s$ when $\boldsymbol{\epsilon}^s = \mathbf{0}$ (no parametric changes) [3]. The motivation behind this choice was the requirement that $\boldsymbol{\Psi}$ be an orthogonal matrix, i.e., $\boldsymbol{\Psi}^T\boldsymbol{\Psi} = \mathbf{I}$. The issue with this, however, is that the square root of $\mathbf{M}^0_{\mathtt{II}}$ must be computed which usually requires a certain amount of time. Fortunately the following simple expression of $\boldsymbol{\Psi}$ can also be used:

$$\boldsymbol{\Psi} = \mathbf{M}^0_{\mathtt{II}}\widetilde{\mathbf{X}}^0. \tag{11}$$

To derive Eq. (10), a MAC-based correlation criterion can be invoked, see [22]. This yields $\boldsymbol{\Psi}^T\widehat{\mathbf{X}}_p = \mathbf{I}$ which can be easily checked by considering Eq. (10). A theoretical derivation of the proposed correlation criterion for obtaining $\boldsymbol{\Psi}$ is given in Appendix A. Thus the following alternative expressions for the transformation matrix and the reduced substructure matrices at the interpolation points are considered:

$$\widehat{\mathbf{T}}_p = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ (\mathbf{X}_{\mathtt{st}})_p & \widehat{\mathbf{X}}_p \end{bmatrix} \quad , \quad \widehat{\mathbf{M}}_p = \widehat{\mathbf{T}}_p^T\mathbf{M}(\boldsymbol{\epsilon}_p)\widehat{\mathbf{T}}_p \quad , \quad \widehat{\mathbf{K}}_p = \widehat{\mathbf{T}}_p^T\mathbf{K}(\boldsymbol{\epsilon}_p)\widehat{\mathbf{T}}_p. \tag{12}$$

The determination of the interpolated reduced mass and stiffness matrices of the substructures follows from classic interpolation, e.g., using Lagrangian interpolation functions $N_p(\boldsymbol{\xi}^s)$ with parametric coordinates $\boldsymbol{\xi}^s$ defined so that they are equal to one or zero at some points $\boldsymbol{\xi}_p$. These points coincide with the "physical" interpolation points $\boldsymbol{\epsilon}_p$ and are here chosen as Gauss points. To fix ideas, for a 1D parametric space $\epsilon \in [-\delta\,,\,\delta]$, a quadratic interpolation with three points ϵ_p ($p = 1, 2, 3$) means selecting these points as $-\epsilon_1 = \epsilon_3 = \sqrt{3/5}\,\delta$ and $\epsilon_2 = 0$, and choosing as parametric coordinate $\xi = \sqrt{5/3}\,(\epsilon/\delta)$ with $\xi \in [-\sqrt{5/3}\,,\,\sqrt{5/3}]$ and interpolation points $-\xi_1 = \xi_3 = 1$ and $\xi_2 = 0$. To sum up, the reduced substructure matrices can be approximated as follows:

$$\widehat{\mathbf{M}}(\boldsymbol{\epsilon}^s) = \widehat{\mathbf{M}}(\boldsymbol{\xi}^s) \approx \sum_{p=1}^{n^p} N_p(\boldsymbol{\xi}^s)\widehat{\mathbf{M}}_p \quad , \quad \widehat{\mathbf{K}}(\boldsymbol{\epsilon}^s) = \widehat{\mathbf{K}}(\boldsymbol{\xi}^s) \approx \sum_{p=1}^{n^p} N_p(\boldsymbol{\xi}^s)\widehat{\mathbf{K}}_p. \tag{13}$$

The interest behind this interpolation strategy is to compute a small number n^p of matrices $\widehat{\mathbf{M}}_p$ and $\widehat{\mathbf{K}}_p$. These matrices are identical for all the substructures, and therefore they only need to be computed once. The rest of the computational load concerns the computation of $\widehat{\mathbf{M}}(\boldsymbol{\epsilon}^s)$ and $\widehat{\mathbf{K}}(\boldsymbol{\epsilon}^s)$ for any set of parameters $\boldsymbol{\epsilon}^s$ via interpolation, which is low-time consuming.

Remark. In [17], a 2D parametric space combined with an eight-point integration scheme that uses Serendipity functions and Chebyshev (interpolation) points was considered. Instead, the choice is made here to use Lagrangian polynomials with Gauss points. This seems to be a good and simple alternative to the analysis of models with an arbitrary number of parameters.

2.2. Reduced modeling of nearly periodic structures

The reduced dynamic equation of a substructure s with interpolated reduced matrices $\widehat{\mathbf{M}}(\boldsymbol{\epsilon}^s)$ and $\widehat{\mathbf{K}}(\boldsymbol{\epsilon}^s)$ can be written as:

$$\left(-\omega^2\widehat{\mathbf{M}}(\boldsymbol{\epsilon}^s) + \mathtt{i}\omega\widehat{\mathbf{C}}(\boldsymbol{\epsilon}^s) + \widehat{\mathbf{K}}(\boldsymbol{\epsilon}^s)\right)\begin{bmatrix}\widehat{\mathbf{u}}^s_{\mathtt{B}} \\ \widehat{\boldsymbol{\alpha}}^s\end{bmatrix} = \begin{bmatrix}\mathbf{F}^s_{\mathtt{B}} \\ \mathbf{0}\end{bmatrix}, \tag{14}$$

[3] Note that, in [17], $\boldsymbol{\Psi} = \left((\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\right)^T\widetilde{\mathbf{X}}^0$ which is equivalent to $\boldsymbol{\Psi} = (\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\widetilde{\mathbf{X}}^0$ since $(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}$ is a symmetric matrix (this can be explained since $\mathbf{M}^0_{\mathtt{II}}$ is symmetric and positive definite).

with $\widehat{\mathbf{C}}(\boldsymbol{\epsilon}^s) = a\widehat{\mathbf{M}}(\boldsymbol{\epsilon}^s) + b\widehat{\mathbf{K}}(\boldsymbol{\epsilon}^s)$ the damping matrix (see Eq. (4)), $\widehat{\mathbf{u}}_{\mathtt{B}}^s$ the vector of boundary displacements [4], $\widehat{\boldsymbol{\alpha}}^s$ the vector of generalized coordinates (fixed interface modes), and $\mathbf{F}_{\mathtt{B}}^s$ the vector of boundary forces (no internal forces). The dynamic equation of a whole nearly periodic structure composed of n^s substructures with different sets of parameters $\boldsymbol{\epsilon}^s$ ($s = 1, \ldots, n^s$) follows from classic matrix assembly. This yields:

$$\left(-\omega^2\widehat{\mathbf{M}}_{\mathtt{a}}(\boldsymbol{\epsilon}_{\mathtt{a}}) + \mathtt{i}\omega\widehat{\mathbf{C}}_{\mathtt{a}}(\boldsymbol{\epsilon}_{\mathtt{a}}) + \widehat{\mathbf{K}}_{\mathtt{a}}(\boldsymbol{\epsilon}_{\mathtt{a}})\right)\begin{bmatrix}(\widehat{\mathbf{u}}_{\mathtt{B}})_{\mathtt{a}}\\ \widehat{\boldsymbol{\alpha}}_{\mathtt{a}}\end{bmatrix} = \begin{bmatrix}(\mathbf{F}_{\mathtt{B}})_{\mathtt{a}}\\ \mathbf{0}\end{bmatrix}, \tag{15}$$

where $\boldsymbol{\epsilon}_{\mathtt{a}} = [(\boldsymbol{\epsilon}^1)^T \cdots (\boldsymbol{\epsilon}^{n^s})^T]^T$ and $\widehat{\boldsymbol{\alpha}}_{\mathtt{a}} = [(\widehat{\boldsymbol{\alpha}}^1)^T \cdots (\widehat{\boldsymbol{\alpha}}^{n^s})^T]^T$. Here, $\widehat{\mathbf{M}}_{\mathtt{a}}(\boldsymbol{\epsilon}_{\mathtt{a}})$, $\widehat{\mathbf{C}}_{\mathtt{a}}(\boldsymbol{\epsilon}_{\mathtt{a}})$ and $\widehat{\mathbf{K}}_{\mathtt{a}}(\boldsymbol{\epsilon}_{\mathtt{a}})$ denote the reduced mass, damping and stiffness matrices of the structure, $(\widehat{\mathbf{u}}_{\mathtt{B}})_{\mathtt{a}}$ and $(\mathbf{F}_{\mathtt{B}})_{\mathtt{a}}$ denote the displacement and force vectors (respectively) at the interface between the substructures (the interface being defined as the union of the substructure boundaries). Kinematic constraints for some interface DOFs can be simply applied in a standard way by removing the related rows/columns in the matrix equation (15).

In Eq. (15), the number of components of the vector $\widehat{\boldsymbol{\alpha}}_{\mathtt{a}}$ is equal to the number of substructures times the number $M_{\mathtt{I}}$ of fixed interface modes retained per substructure (a few modes). On the other hand, $(\widehat{\mathbf{u}}_{\mathtt{B}})_{\mathtt{a}}$ refers to the interface DOFs whose number (say, $N_{\mathtt{a}}$) is usually large. This can strongly affect the numerical efficiency of the proposed strategy. To address this issue, an interface reduction which uses the interface modes of the equivalent purely periodic structure (matrix $\widetilde{\mathbf{X}}_{\mathtt{a}}^0$), where a reduced number $M_{\mathtt{a}} \ll N_{\mathtt{a}}$ of these modes are retained, has been proposed in [17]. These modes represent the eigenvectors of the matrix pencil $((\widehat{\mathbf{K}}_{\mathtt{BB}}^0)_{\mathtt{a}}, (\widehat{\mathbf{M}}_{\mathtt{BB}}^0)_{\mathtt{a}})$ where subscript $\mathtt{B}$ denotes interface DOFs and superscript 0 means that $\boldsymbol{\epsilon} = \mathbf{0}$. They can be classically normalized in such a way that $(\widetilde{\mathbf{X}}_{\mathtt{a}}^0)^T(\widehat{\mathbf{M}}_{\mathtt{BB}}^0)_{\mathtt{a}}\widetilde{\mathbf{X}}_{\mathtt{a}}^0 = \mathbf{I}$ and $(\widetilde{\mathbf{X}}_{\mathtt{a}}^0)^T(\widehat{\mathbf{K}}_{\mathtt{BB}}^0)_{\mathtt{a}}\widetilde{\mathbf{X}}_{\mathtt{a}}^0 = \widetilde{\boldsymbol{\Lambda}}_{\mathtt{a}}^0$ with $\widetilde{\boldsymbol{\Lambda}}_{\mathtt{a}}^0$ the diagonal matrix of the retained eigenvalues of $((\widehat{\mathbf{K}}_{\mathtt{BB}}^0)_{\mathtt{a}}, (\widehat{\mathbf{M}}_{\mathtt{BB}}^0)_{\mathtt{a}})$. In the present study, some improvement of the interface reduction technique is proposed as follows. The idea is to enrich the mode basis with residual flexibility vectors in the same way as in the framework of free-interface methods [9, 30]. For the sake of simplicity, let us assume that $(\widehat{\mathbf{K}}_{\mathtt{BB}}^0)_{\mathtt{a}}$ has full rank which means that a sufficient number of kinematic constraints are considered to prevent rigid body modes. Thus it is proposed to approximate the displacement vector $(\widehat{\mathbf{u}}_{\mathtt{B}})_{\mathtt{a}}$ as follows:

$$(\widehat{\mathbf{u}}_{\mathtt{B}})_{\mathtt{a}} \approx \widetilde{\mathbf{X}}_{\mathtt{a}}^0\widetilde{\boldsymbol{\beta}}_{\mathtt{a}} + \widetilde{\mathbf{G}}_{\mathtt{a}}^0(\mathbf{L}_{\mathtt{ex}})_{\mathtt{a}}(\mathbf{F}_{\mathtt{B}})_{\mathtt{a}}, \tag{16}$$

with $\widetilde{\boldsymbol{\beta}}_{\mathtt{a}}$ the vector of generalized coordinates for the interface modes $\widetilde{\mathbf{X}}_{\mathtt{a}}^0$ and $(\mathbf{L}_{\mathtt{ex}})_{\mathtt{a}}$ a Boolean matrix to localize the DOFs where external forces (excitation) are applied. Also, $\widetilde{\mathbf{G}}_{\mathtt{a}}^0$ is the residual flexibility matrix defined by:

$$\widetilde{\mathbf{G}}_{\mathtt{a}}^0 = \left((\widehat{\mathbf{K}}_{\mathtt{BB}}^0)_{\mathtt{a}}^{-1} - \widetilde{\mathbf{X}}_{\mathtt{a}}^0(\widetilde{\boldsymbol{\Lambda}}_{\mathtt{a}}^0)^{-1}(\widetilde{\mathbf{X}}_{\mathtt{a}}^0)^T\right)(\mathbf{L}_{\mathtt{ex}})_{\mathtt{a}}^T. \tag{17}$$

Left multiplying Eq. (16) by $(\mathbf{L}_{\mathtt{ex}})_{\mathtt{a}}$ gives:

$$(\widehat{\mathbf{u}}_{\mathtt{ex}})_{\mathtt{a}} = (\mathbf{L}_{\mathtt{ex}})_{\mathtt{a}}\widetilde{\mathbf{X}}_{\mathtt{a}}^0\widetilde{\boldsymbol{\beta}}_{\mathtt{a}} + (\mathbf{L}_{\mathtt{ex}})_{\mathtt{a}}\widetilde{\mathbf{G}}_{\mathtt{a}}^0(\mathbf{L}_{\mathtt{ex}})_{\mathtt{a}}(\mathbf{F}_{\mathtt{B}}^s)_{\mathtt{a}}, \tag{18}$$

where $(\widehat{\mathbf{u}}_{\mathtt{ex}})_{\mathtt{a}} \approx (\mathbf{L}_{\mathtt{ex}})_{\mathtt{a}}(\widehat{\mathbf{u}}_{\mathtt{B}})_{\mathtt{a}}$ is the displacement vector for the excitation DOFs. Eq. (18) yields:

$$(\mathbf{L}_{\mathtt{ex}})_{\mathtt{a}}(\mathbf{F}_{\mathtt{B}}^s)_{\mathtt{a}} = (\widetilde{\boldsymbol{\Theta}}_{\mathtt{ex}}^0)_{\mathtt{a}}\left((\widehat{\mathbf{u}}_{\mathtt{ex}})_{\mathtt{a}} - (\mathbf{L}_{\mathtt{ex}})_{\mathtt{a}}\widetilde{\mathbf{X}}_{\mathtt{a}}^0\widetilde{\boldsymbol{\beta}}_{\mathtt{a}}\right) \qquad \text{where} \qquad (\widetilde{\boldsymbol{\Theta}}_{\mathtt{ex}}^0)_{\mathtt{a}} = \left((\mathbf{L}_{\mathtt{ex}})_{\mathtt{a}}\widetilde{\mathbf{G}}_{\mathtt{a}}^0\right)^{-1}, \tag{19}$$

[4] $\widehat{\mathbf{u}}_{\mathtt{B}}^s$ is computed using the interpolation strategy, and therefore it represents an approximate of the displacement vector $\mathbf{u}_{\mathtt{B}}^s$.

with $(\widetilde{\boldsymbol{\Theta}}_{\mathtt{ex}})^0_{\mathtt{a}}$ a square matrix whose size corresponds to the number of excitation DOFs (usually a small number). Inserting Eq. (19) into Eq. (16) finally gives:

$$(\widehat{\mathbf{u}}_{\mathtt{B}})_{\mathtt{a}} \approx \left[\widetilde{\mathbf{X}}^0_{\mathtt{a}} - \widetilde{\mathbf{G}}^0_{\mathtt{a}}(\widetilde{\boldsymbol{\Theta}}^0_{\mathtt{ex}})_{\mathtt{a}}(\mathbf{L}_{\mathtt{ex}})_{\mathtt{a}}\widetilde{\mathbf{X}}^0_{\mathtt{a}}\right]\widetilde{\boldsymbol{\beta}}_{\mathtt{a}} + \left[\widetilde{\mathbf{G}}^0_{\mathtt{a}}(\widetilde{\boldsymbol{\Theta}}^0_{\mathtt{ex}})_{\mathtt{a}}\right](\widehat{\mathbf{u}}_{\mathtt{ex}})_{\mathtt{a}}. \tag{20}$$

The proposed interface reduction is provided by Eq. (20). As a result, the following approximation for the vector of interface displacements and the vector of generalized coordinates can be proposed:

$$\begin{bmatrix}(\widehat{\mathbf{u}}_{\mathtt{B}})_{\mathtt{a}}\\ \widehat{\boldsymbol{\alpha}}_{\mathtt{a}}\end{bmatrix} \approx \begin{bmatrix}\widetilde{\mathbf{X}}^0_{\mathtt{a}} - \widetilde{\mathbf{G}}^0_{\mathtt{a}}(\widetilde{\boldsymbol{\Theta}}^0_{\mathtt{ex}})_{\mathtt{a}}(\mathbf{L}_{\mathtt{ex}})_{\mathtt{a}}\widetilde{\mathbf{X}}^0_{\mathtt{a}} & \widetilde{\mathbf{G}}^0_{\mathtt{a}}(\widetilde{\boldsymbol{\Theta}}^0_{\mathtt{ex}})_{\mathtt{a}} & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{I}\end{bmatrix}\begin{bmatrix}\widetilde{\boldsymbol{\beta}}_{\mathtt{a}}\\ (\widehat{\mathbf{u}}_{\mathtt{ex}})_{\mathtt{a}}\\ \widehat{\boldsymbol{\alpha}}_{\mathtt{a}}\end{bmatrix} = \widetilde{\mathbf{T}}^0_{\mathtt{a}}\begin{bmatrix}\widetilde{\boldsymbol{\beta}}_{\mathtt{a}}\\ (\widehat{\mathbf{u}}_{\mathtt{ex}})_{\mathtt{a}}\\ \widehat{\boldsymbol{\alpha}}_{\mathtt{a}}\end{bmatrix}. \tag{21}$$

For the sake of conciseness, this equation can be rewritten as follows:

$$\begin{bmatrix}(\widehat{\mathbf{u}}_{\mathtt{B}})_{\mathtt{a}}\\ \widehat{\boldsymbol{\alpha}}_{\mathtt{a}}\end{bmatrix} \approx \begin{bmatrix}\widetilde{\mathbf{X}}^0_{\mathtt{a}} & \mathbf{0}\\ \mathbf{0} & \mathbf{I}\end{bmatrix}\begin{bmatrix}\widetilde{\boldsymbol{\beta}}_{\mathtt{a}}\\ \widehat{\boldsymbol{\alpha}}_{\mathtt{a}}\end{bmatrix} = \widetilde{\mathbf{T}}^0_{\mathtt{a}}\begin{bmatrix}\widetilde{\boldsymbol{\beta}}_{\mathtt{a}}\\ \widehat{\boldsymbol{\alpha}}_{\mathtt{a}}\end{bmatrix}, \tag{22}$$

with $\widetilde{\mathbf{X}}^0_{\mathtt{a}}$ the updated matrix of interface modes, and $\widetilde{\boldsymbol{\beta}}_{\mathtt{a}}$ the updated vector of generalized coordinates:

$$\widetilde{\mathbf{X}}^0_{\mathtt{a}} \leftarrow \left[\widetilde{\mathbf{X}}^0_{\mathtt{a}} - \widetilde{\mathbf{G}}^0_{\mathtt{a}}(\widetilde{\boldsymbol{\Theta}}^0_{\mathtt{ex}})_{\mathtt{a}}(\mathbf{L}_{\mathtt{ex}})_{\mathtt{a}}\widetilde{\mathbf{X}}^0_{\mathtt{a}} \quad \widetilde{\mathbf{G}}^0_{\mathtt{a}}(\widetilde{\boldsymbol{\Theta}}^0_{\mathtt{ex}})_{\mathtt{a}}\right] \quad \text{and} \quad \widetilde{\boldsymbol{\beta}}_{\mathtt{a}} \leftarrow \begin{bmatrix}\widetilde{\boldsymbol{\beta}}_{\mathtt{a}}\\ (\widehat{\mathbf{u}}_{\mathtt{ex}})_{\mathtt{a}}\end{bmatrix}. \tag{23}$$

The related reduced dynamic equation of the structure writes:

$$\left(-\omega^2\mathbb{M}_{\mathtt{a}}(\boldsymbol{\epsilon}_{\mathtt{a}}) + \mathtt{i}\omega\mathbb{C}_{\mathtt{a}}(\boldsymbol{\epsilon}_{\mathtt{a}}) + \mathbb{K}_{\mathtt{a}}(\boldsymbol{\epsilon}_{\mathtt{a}})\right)\begin{bmatrix}\widetilde{\boldsymbol{\beta}}_{\mathtt{a}}\\ \widehat{\boldsymbol{\alpha}}_{\mathtt{a}}\end{bmatrix} = \mathbb{F}_{\mathtt{a}}, \tag{24}$$

where

$$\mathbb{M}_{\mathtt{a}}(\boldsymbol{\epsilon}_{\mathtt{a}}) = (\widetilde{\mathbf{T}}^0_{\mathtt{a}})^T\widehat{\mathbf{M}}_{\mathtt{a}}(\boldsymbol{\epsilon}_{\mathtt{a}})\widetilde{\mathbf{T}}^0_{\mathtt{a}}, \tag{25}$$

$$\mathbb{C}_{\mathtt{a}}(\boldsymbol{\epsilon}_{\mathtt{a}}) = (\widetilde{\mathbf{T}}^0_{\mathtt{a}})^T\widehat{\mathbf{C}}_{\mathtt{a}}(\boldsymbol{\epsilon}_{\mathtt{a}})\widetilde{\mathbf{T}}^0_{\mathtt{a}}, \tag{26}$$

$$\mathbb{K}_{\mathtt{a}}(\boldsymbol{\epsilon}_{\mathtt{a}}) = (\widetilde{\mathbf{T}}^0_{\mathtt{a}})^T\widehat{\mathbf{K}}_{\mathtt{a}}(\boldsymbol{\epsilon}_{\mathtt{a}})\widetilde{\mathbf{T}}^0_{\mathtt{a}}, \tag{27}$$

$$\mathbb{F}_{\mathtt{a}} = (\widetilde{\mathbf{T}}^0_{\mathtt{a}})^T\begin{bmatrix}(\mathbf{F}_{\mathtt{B}})_{\mathtt{a}}\\ \mathbf{0}\end{bmatrix}. \tag{28}$$

Solving Eq. (24) yields the vectors $\widetilde{\boldsymbol{\beta}}_{\mathtt{a}}$ and $\widehat{\boldsymbol{\alpha}}_{\mathtt{a}}$. Subsequently the vector of interface displacements $(\widehat{\mathbf{u}}_{\mathtt{B}})_{\mathtt{a}}$ can be determined as $(\widehat{\mathbf{u}}_{\mathtt{B}})_{\mathtt{a}} \approx \widetilde{\mathbf{X}}^0_{\mathtt{a}}\widetilde{\boldsymbol{\beta}}_{\mathtt{a}}$, see Eq. (20). Besides an approximate of the displacement vector $\mathbf{u}^s_{\mathtt{I}}$ of the internal nodes of the substructures can be determined by considering the local vectors $\widehat{\mathbf{u}}^s_{\mathtt{B}}$ and $\widehat{\boldsymbol{\alpha}}^s$ — which can be easily obtained from the global vectors $(\widehat{\mathbf{u}}_{\mathtt{B}})_{\mathtt{a}}$ and $\widehat{\boldsymbol{\alpha}}_{\mathtt{a}}$ using localization matrices — and the transformation matrix $\widehat{\mathbf{T}}^0$ of the substructures when $\boldsymbol{\epsilon}^s = \mathbf{0}$ as explained in [17].

3. Model reduction with basis enrichment

The interpolation strategy for modeling nearly periodic structures has been recalled and reviewed in Sec. 2. The strategy consists in (i) interpolating reduced substructure matrices between some interpolation points and (ii) reducing the number of interface DOFs, between the substructures, using interface modes. Although efficient in terms of computational saving, this procedure is however prone to numerical limitations and lack of accuracy, especially at high frequencies, for the following reasons:

- In Eq. (10), it is implicitly assumed that the matrices $\boldsymbol{\Psi}^T\widetilde{\mathbf{X}}_p$ ($p = 1, \ldots, n^p$) are invertible, i.e., that they have full rank. This means that the column ranges of the matrices $\widetilde{\mathbf{X}}_p$ have to be "sufficiently close" to the column range of the matrix $\boldsymbol{\Psi}$ as expressed in Eq. (11). From the practical point of view, this requires considering dispersion parameters δ_{xk}, δ_{yk} and $\delta_{\theta k}$ (see comments about Eqs. (1) and (2)) of reasonable values. This also requires considering a few low-order fixed interface modes in the reduced matrices $\widetilde{\mathbf{X}}_p$. Indeed low-order modes usually exhibit regular shapes which as such are likely to be spanned by the column vectors of $\boldsymbol{\Psi}$, as opposed to high-order modes. This prevents these modes to be taken into account, and therefore the interpolation strategy to be accurate at high frequencies.

- In the interface reduction method proposed in Sec. 2.2, the reduced matrix of interface modes $\widetilde{\mathbf{X}}^0_{\mathtt{a}}$ of an equivalent purely periodic structures is used. This choice is motivated by the need of computing these modes only once, i.e., to avoid computing them for reanalysing nearly periodic structures with different sets of parameters $\boldsymbol{\epsilon}_{\mathtt{a}}$. Since these interface modes are different from the true ones — i.e., those of a nearly periodic structure with parametric changes –, the convergence of this mode basis cannot be guaranteed.

To solve these issues, two basis enrichment techniques are proposed as discussed hereafter.

3.1. High-order static modes

To improve the accuracy of the interpolation strategy, high-order static correction terms can be considered to enrich the CB basis of the substructures. These correction terms represent the static contributions of the residual fixed interface modes which are not retained in the CB basis. Clearly speaking, let $(\mathbf{u}^s_{\mathtt{I}})_{\mathtt{r}}$ be the displacement vector associated with the residual fixed interface modes. Then it can be proven (see Appendix B) that $(\mathbf{u}^s_{\mathtt{I}})_{\mathtt{r}} \approx \omega^2\mathbf{X}^{\mathtt{cor}}_{\mathtt{st}}\mathbf{u}^s_{\mathtt{B}}$ with $\mathbf{X}^{\mathtt{cor}}_{\mathtt{st}}$ a matrix of high-order static modes given by [23, 24]:

$$\mathbf{X}^{\mathtt{cor}}_{\mathtt{st}} = \left(\mathbf{K}^{-1}_{\mathtt{II}} - \widetilde{\mathbf{X}}\widetilde{\boldsymbol{\Lambda}}^{-1}\widetilde{\mathbf{X}}^T\right)\left(\mathbf{M}_{\mathtt{II}}\mathbf{X}_{\mathtt{st}} + \mathbf{M}_{\mathtt{IB}}\right), \tag{29}$$

where the dependency on parameters $\boldsymbol{\epsilon}^s$ has been dropped for the sake of conciseness. Here, $\widetilde{\mathbf{X}}$ and $\widetilde{\boldsymbol{\Lambda}}$ denote, respectively, the reduced matrix of fixed interface modes (see Sec. 2.1) and the related reduced diagonal matrix of eigenvalues. The fact that $(\mathbf{u}^s_{\mathtt{I}})_{\mathtt{r}} \approx \omega^2\mathbf{X}^{\mathtt{cor}}_{\mathtt{st}}\mathbf{u}^s_{\mathtt{B}}$ means that $(\mathbf{u}^s_{\mathtt{I}})_{\mathtt{r}}$ is located in the column space of $\mathbf{X}^{\mathtt{cor}}_{\mathtt{st}}$. In this sense, the column vectors of $\mathbf{X}^{\mathtt{cor}}_{\mathtt{st}}$ can be used to enrich the CB basis. Note that the matrix $\mathbf{X}^{\mathtt{cor}}_{\mathtt{st}}$ is usually badly conditioned or rank-deficient because of the fact that part of the high-order static modes are nearly collinear. To solve this issue, the following strategy can be considered. First the singular value decomposition (SVD) of $\mathbf{X}^{\mathtt{cor}}_{\mathtt{st}}$ is invoked:

$$\mathbf{X}^{\mathtt{cor}}_{\mathtt{st}} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T, \tag{30}$$

with $\boldsymbol{\Sigma}$ the matrix of singular values and $\mathbf{U}$ and $\mathbf{V}$ the related orthogonal matrices of left and right singular vectors, respectively. By retaining only a few largest singular values of $\mathbf{X}^{\mathtt{cor}}_{\mathtt{st}}$, i.e., by considering a reduced matrix $\widetilde{\boldsymbol{\Sigma}}$ with $M^{\mathtt{e}}_{\mathtt{st}}$ singular values, the following approximation can be proposed:

$$\mathbf{X}^{\mathtt{cor}}_{\mathtt{st}} \approx \widetilde{\mathbf{U}}\widetilde{\boldsymbol{\Sigma}}\widetilde{\mathbf{V}}^T, \tag{31}$$

where $\widetilde{\mathbf{U}}$ and $\widetilde{\mathbf{V}}$ are reduced matrices of singular vectors. From Eq. (31), it appears that the column vectors of $\widetilde{\mathbf{U}}$ constitute a relevant basis for the column space of $\mathbf{X}^{\mathtt{cor}}_{\mathtt{st}}$. In this sense this vector basis can be used to enrich the CB basis of the substructures. Following the interpolation strategy proposed in Sec. 2.1, the reduced matrices of left singular vectors $\widetilde{\mathbf{U}}_p$ at interpolation points "$\boldsymbol{\epsilon}_p$" need to be expressed

in compatible coordinate systems in the same way as in Eq. (10). In this sense, the following alternative reduced matrices of high-order static modes are considered:

$$\widehat{\mathbf{U}}_p = \widetilde{\mathbf{U}}_p \left((\widetilde{\mathbf{U}}^0)^T \widetilde{\mathbf{U}}_p \right)^{-1}, \tag{32}$$

where $\widetilde{\mathbf{U}}_p$ and $\widetilde{\mathbf{U}}^0$ represent $\widetilde{\mathbf{U}}$ when $\boldsymbol{\epsilon}^s = \boldsymbol{\epsilon}_p$ and $\boldsymbol{\epsilon}^s = \mathbf{0}$, respectively. As a result, the transformation matrices and the reduced substructure matrices at the interpolation points become, see Eq. (12):

$$\widehat{\mathbf{T}}_p = \begin{bmatrix} \mathbf{I} & \mathbf{0} & \mathbf{0} \\ (\mathbf{X}_{\mathtt{st}})_p & \widehat{\mathbf{X}}_p & \widehat{\mathbf{U}}_p \end{bmatrix} \quad , \quad \widehat{\mathbf{M}}_p = \widehat{\mathbf{T}}_p^T \mathbf{M}(\boldsymbol{\epsilon}_p) \widehat{\mathbf{T}}_p \quad , \quad \widehat{\mathbf{K}}_p = \widehat{\mathbf{T}}_p^T \mathbf{K}(\boldsymbol{\epsilon}_p) \widehat{\mathbf{T}}_p. \tag{33}$$

The approximation of the reduced substructure matrices for arbitrary sets of parameters $\boldsymbol{\epsilon}^s$ follows from interpolation, see Eq. (13). The reduced dynamic equation of a substructure s can be written as in Eq. (14) with $\widehat{\boldsymbol{\alpha}}^s$ the $(M_{\mathtt{I}} + M_{\mathtt{st}}^{\mathtt{e}}) \times 1$ vector of generalized coordinates for the fixed interface modes and high-order static modes.

3.2. Static correction of interface modes

To improve the convergence of the basis of interface modes (matrix $\widetilde{\mathbf{X}}_{\mathtt{a}}^0$), the following strategy can be proposed. Let us first consider the following matrix of residual force vectors resulting from parametric changes [25, 26]:

$$\Delta(\mathbf{F}_{\mathtt{B}})_{\mathtt{a}} = -\Delta(\widehat{\mathbf{K}}_{\mathtt{BB}})_{\mathtt{a}} \, \widetilde{\mathbf{X}}_{\mathtt{a}}^0. \tag{34}$$

Here, $\Delta(\widehat{\mathbf{K}}_{\mathtt{BB}})_{\mathtt{a}}$ denotes the perturbation of the stiffness matrix (interface DOFs) due to parametric changes:

$$\Delta(\widehat{\mathbf{K}}_{\mathtt{BB}})_{\mathtt{a}} = (\widehat{\mathbf{K}}_{\mathtt{BB}})_{\mathtt{a}}(\boldsymbol{\epsilon}_{\mathtt{a}}) - (\widehat{\mathbf{K}}_{\mathtt{BB}}^0)_{\mathtt{a}}, \tag{35}$$

with $(\widehat{\mathbf{K}}_{\mathtt{BB}})_{\mathtt{a}}(\boldsymbol{\epsilon}_{\mathtt{a}})$ and $(\widehat{\mathbf{K}}_{\mathtt{BB}}^0)_{\mathtt{a}}$ the stiffness matrices for the interface DOFs of the nearly periodic structure ($\boldsymbol{\epsilon}_{\mathtt{a}} \neq \mathbf{0}$) and the purely periodic one ($\boldsymbol{\epsilon}_{\mathtt{a}} = \mathbf{0}$), respectively. Also, in Eq. (34), $\widetilde{\mathbf{X}}_{\mathtt{a}}^0$ is the reduced matrix of interface modes provided by Eq. (23) which is used here to generate an appropriate space for the interface displacements. From Eq. (34), the following matrix of static correction vectors due to parametric changes can be proposed to enrich the basis of interface modes:

$$\Delta\widetilde{\mathbf{X}}_{\mathtt{a}} = (\widehat{\mathbf{K}}_{\mathtt{BB}}^0)_{\mathtt{a}}^{-1} \, \Delta(\mathbf{F}_{\mathtt{B}})_{\mathtt{a}} = -(\widehat{\mathbf{K}}_{\mathtt{BB}}^0)_{\mathtt{a}}^{-1} \, \Delta(\widehat{\mathbf{K}}_{\mathtt{BB}})_{\mathtt{a}} \, \widetilde{\mathbf{X}}_{\mathtt{a}}^0. \tag{36}$$

For computational purposes, the static correction vectors can be orthogonalized through a SVD:

$$\Delta\widetilde{\mathbf{X}}_{\mathtt{a}} = \mathbf{U}_{\mathtt{a}} \boldsymbol{\Sigma}_{\mathtt{a}} \mathbf{V}_{\mathtt{a}}^T. \tag{37}$$

Thus a matrix with orthogonal vectors $\mathbf{U}_{\mathtt{a}}$, representing the left singular vectors of $\Delta\widetilde{\mathbf{X}}_{\mathtt{a}}$, which span the same vector space as $\Delta\widetilde{\mathbf{X}}_{\mathtt{a}}$ can be obtained and used as an alternative matrix of static correction vectors. Hence, the following updated matrix of interface modes can be considered to improve the accuracy of the proposed approach:

$$\widetilde{\mathbf{X}}_{\mathtt{a}}^0 \leftarrow \left[\widetilde{\mathbf{X}}_{\mathtt{a}}^0 \quad \mathbf{U}_{\mathtt{a}} \right]. \tag{38}$$

Note that the matrix $\mathbf{U}_{\mathtt{a}}$ contains the same number of vectors as $\widetilde{\mathbf{X}}_{\mathtt{a}}^0$, i.e., the updated matrix of interface modes contains twice the number of vectors of the original matrix.

4. Numerical results

The relevance of the interpolation strategy with basis enrichment is demonstrated considering three examples of moderate complexities. Secs. 4.1 and 4.2 focus on the analysis of two nearly periodic structures with 2D substructures under plane stress state. The first example involves substructure models depending on two mesh parameters (mesh distortion); the second one concerns substructure models with three parameters including two mesh parameters. Finally, Sec. 4.3 focuses on the analysis of a nearly periodic structure with 3D substructures where the substructure models depend on three mesh parameters.

4.1. 2D substructures with shape changes

In this first example, a nearly periodic structure under plane stress state composed of 5×3 squared substructures of surface area 0.2×0.2 m^2 and thickness 0.005 m is considered as shown in Fig. 3. These substructures are composed of a stiff core with a soft surrounding layer embedded in a stiff matrix (see Tab. 4.1 for the material properties). Each substructure is meshed with linear triangular elements (three nodes per element and two DOFs per node), providing 1556 internal DOFs and 240 boundary DOFs. Here, the substructure mesh is generated using DistMesh [31]; the related FE model (mass and stiffness matrices) follows from in-house implementations on MATLAB. Geometric changes of the substructures are introduced by moving the positions of the nodes of a baseline substructure — representing a substructure with a circular layer — as follows, see Eq. (1):

$$x_j^s = x_j^0 + \epsilon_1^s f_{x1}(x_j^0, y_j^0) + \epsilon_2^s f_{x2}(x_j^0, y_j^0) \quad , \quad y_j^s = y_j^0 + \epsilon_1^s f_{y1}(x_j^0, y_j^0) + \epsilon_2^s f_{y2}(x_j^0, y_j^0), \tag{39}$$

where shape functions $f_{x1}(x, y)$ and $f_{y1}(x, y)$ are used to generate a homothetic transformation to increase or decrease the inner and outer radii of the circular layer, while shape functions $f_{x2}(x, y)$ and $f_{y2}(x, y)$ are used to generate circumferential ripples of azimuth $\pi/4$. In this case, two mesh parameters ϵ_1^s and ϵ_2^s — representing two independent random variables with different values between the substructures — are considered. The related dispersion parameters are denoted by δ_1 and δ_2 and yield the kind of distorted meshes observed in Fig. 3. Within the framework of the proposed strategy, a 2D interpolation scheme based on 3×3 interpolation points and 9 quadratic Lagrange polynomials $N_p(\xi^s, \eta^s)$ $(p = 1, \ldots, 9)$ with $\xi^s = \sqrt{5/3}(\epsilon_1^s/\delta_1)$ and $\eta^s = \sqrt{5/3}(\epsilon_2^s/\delta_2)$ are considered, see Sec. 2.1. A schematic of the distorted mesh of the substructures at the interpolation points is given in Fig. 4. These interpolation points are numbered using the classic rule for 9-node elements, where point 9 indicates a substructure without distortion.

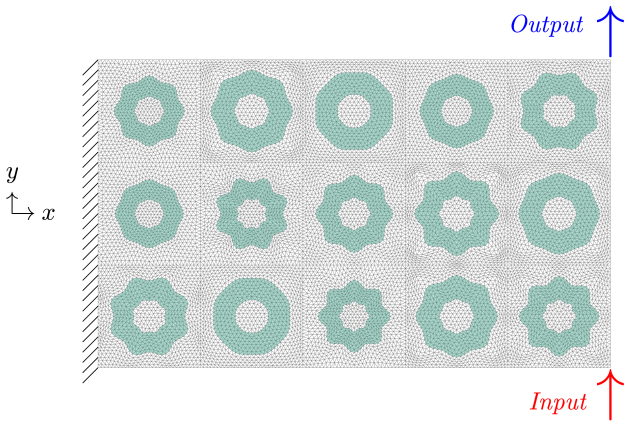


Figure 3: Nearly periodic structure with 2D substructures including circular layers (in green) with shape changes.

Table 1: Material properties of the substructures (first example).

	Density (kg.m^{-3})	Young's modulus (GPa)	Poisson's ratio
First material (gray):	2700	70	0.35
Second material (green):	950	0.15	0.48

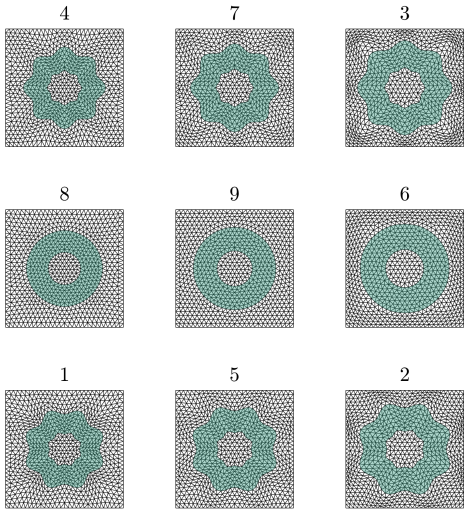


Figure 4: FE meshes of the substructures at interpolation points (first example).

The harmonic response of the nearly periodic structure is analyzed over $[0\,,\,2500]$ Hz. Clamped boundary condition at the left edge and unit harmonic force at the right bottom corner (input) are considered as shown in Fig. 3. Then the quadratic velocity for the vertical direction — i.e., $\omega^2|v_{\mathtt{out}}|^2$ with $v_{\mathtt{out}}$ the $y-$displacement — is recorded at the right top corner (output). FE reference results — i.e., as obtained by simulating the full FE model — for the nearly periodic structure and the equivalent purely periodic one are shown in Fig. 5. To obtain these results, damping coefficients of $a = 0.01$ s^{-1} and $b = 10^{-7}$ s for the substructures are considered, see Eq. (4). The purely periodic structure is a locally resonant metamaterial, which as such possesses band gap properties. The resonant behavior of the substructures is induced by the vibrations of the cores inside the soft elastic layers. For instance, the first three eigenfrequencies of the substructures with clamped boundary are $f_1 \approx f_2 = 1459$ Hz $f_3 = 1728$ Hz as highlighted in Fig. 5 with the red dotted vertical lines. Band gaps occur close to the resonance frequencies of the substructures and yield low vibration levels. Aside from band gaps, the response function of the purely periodic structure exhibits smooth behavior with well-defined resonance peaks over the whole frequency range. On the other hand, the response function of the nearly periodic structure appears to be more irregular with local resonance peaks because of the dissimilarities among the resonance frequencies of the substructures. From Fig. 5, it is clear that the nearly periodic structure behaves in a completely different way than the purely periodic one over the whole frequency range analyzed, which therefore highlights the effect of the parametric (mesh) changes of the substructures.

By using the classic interpolation strategy (Sec. 2), the response function of the nearly periodic structure can be predicted as shown in Fig. 6. In this framework, a CB approximation for the internal DOFs of the substructures with $M_{\mathtt{I}} = 10$ fixed interface modes are considered. The procedure to select these modes consists in retaining those whose eigenfrequencies is less than twice the maximum frequency of interest, i.e., 5000 Hz. The interpolation strategy for the reduced substructure matrices is detailed in Sec. 2.1. Otherwise, a reduction of the number of interface DOFs, between the substructures, is proposed as explained in Sec. 2.2. In this case, $50+1$ modes — representing the classic interface modes

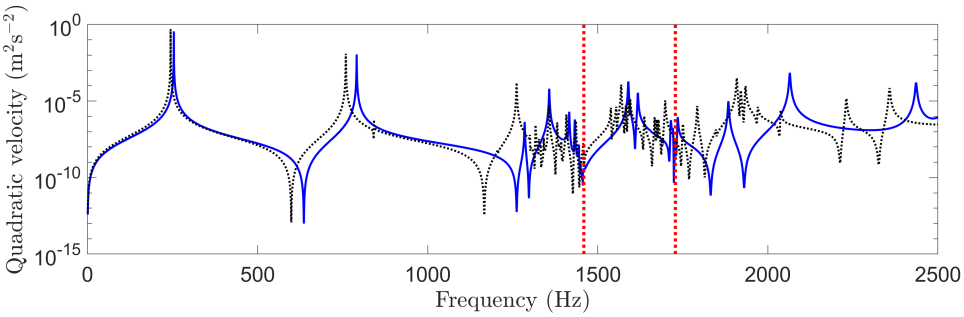


Figure 5: Frequency responses (first example): reference solutions for the purely periodic structure (blue solid line) and the nearly periodic structure (black dotted line). Red dotted vertical lines indicate the eigenfrequencies of the baseline substructure with clamped boundary.

and one additional residual term corresponding to the excitation DOF, see Eq. (23) — are considered. Fig. 6 shows a comparison between the proposed solution and the reference one. An insight into the high frequency range $[1500\,,\,2500]$ Hz is also proposed (see bottom figure) where it becomes clear that the interpolation strategy suffers from a lack of accuracy in this range.

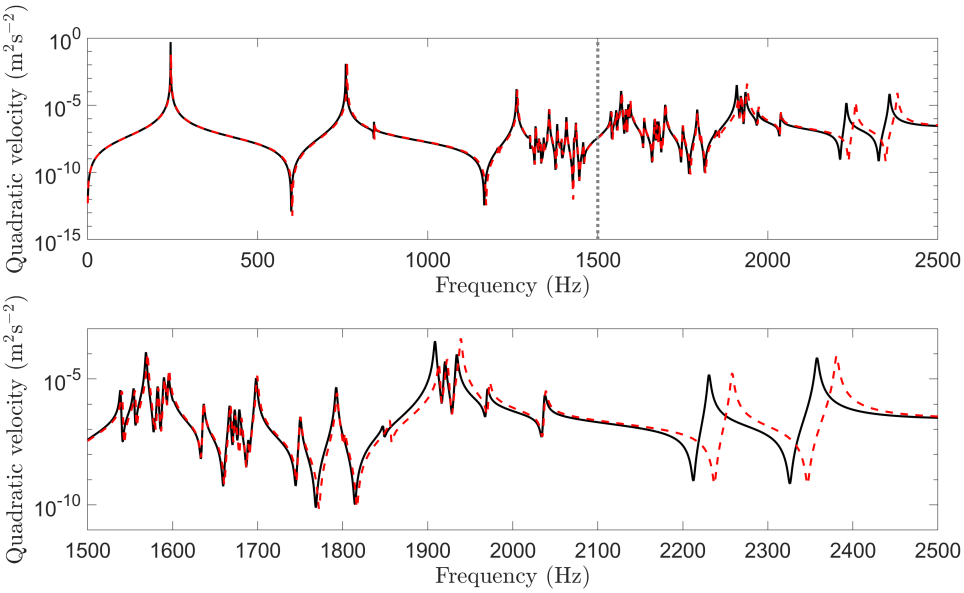


Figure 6: Frequency responses of the nearly periodic structure (first example): reference solution (black solid line) and proposed solution without basis enrichment (red dashed line) over $[0\,,\,2500]$ Hz (top) and $[1500\,,\,2500]$ Hz (bottom).

To address the lack of accuracy of the interpolation strategy, the basis enrichment techniques proposed in Secs. 3.1 and 3.2 are considered. In this case, the CB basis of the substructures at the interpolation points is enriched with $M^{\mathtt{e}}_{\mathtt{st}} = 20$ high-order static modes (matrices $\widehat{\mathbf{U}}_p$), see Eq. (33). In addition, the interface modes of the structure are enriched with static correction vectors (matrix $\mathbf{U}_{\mathtt{a}}$), see Eq. (38). Note that, by definition, the number of these static correction vectors corresponds to the number of interface modes, i.e., 51. By considering the interpolation strategy with basis enrichment, the frequency response can be accurately predicted over the whole frequency range as shown in Fig. 7, even at high frequencies (as expected). In terms of computational saving, both approaches (without or with basis enrichment) yield good time reduction, i.e., more than 90% saving for computing the dynamic equation of the structure. For the sake of clarity, the sizes of the full and reduced models considered are

detailed in Tab. 2.

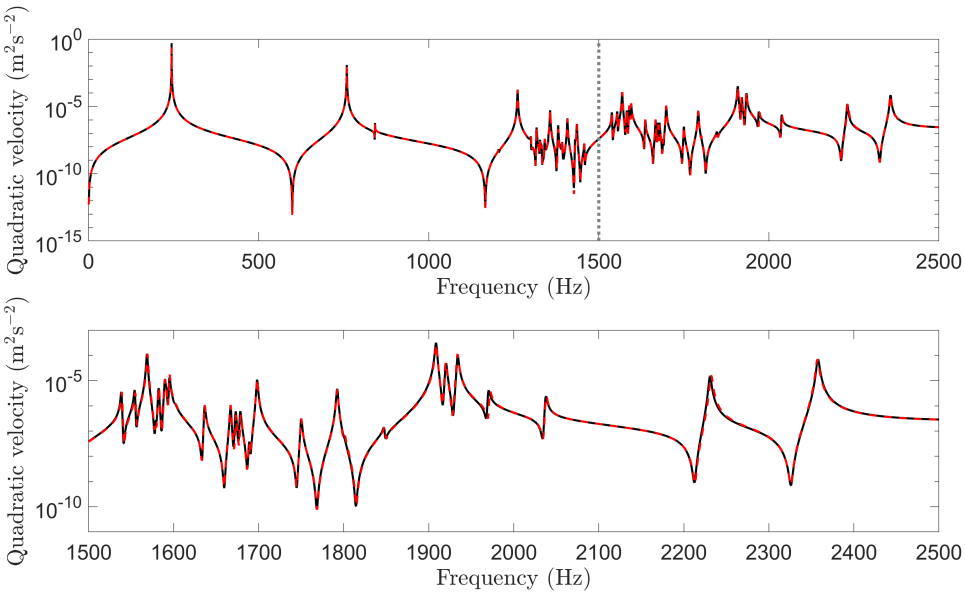


Figure 7: Frequency responses of the nearly periodic structure (first example): reference solution (black solid line) and proposed solution with basis enrichment (red dashed line) over $[0\,,\,2500]$ Hz (top) and $[1500\,,\,2500]$ Hz (bottom).

Table 2: Number of DOFs involved in the models (first example).

	Reference	Interpolated	Interpolated with basis enrichment
Substructure	Boundary: 240 Internal: 1556	Static modes: 240 Fixed interface modes: 10	Static modes: 240 Fixed interface modes: 10 High-order static modes: 20
Interface	2070	Interface modes: 51	Interface modes: 51 Static correction vectors: 51
Structure	$25,410$	$51+15\times 10=201$	$102+15\times(10+20)=552$

4.2. 2D substructures including circular cores with varying positions and thicknesses

Again, a nearly periodic structure under plane stress state with 5×3 substructures (surface area of 0.2×0.2 m^2) is considered as shown in Fig. 8. In this case, each substructure consists of a squared matrix of thickness $h_1 = 0.003$ m with a heavy circular core of thickness $h_2 = 0.005$ m (material properties are given in Tab. 3). Each substructure is meshed with linear triangular elements providing 1452 internal DOFs and 216 boundary DOFs. The related FE mesh and FE model are generated and built using DistMesh and MATLAB. Mesh distortion of the substructures consists in moving the positions of the core in the (x,y) plane, which can be done by considering two shape functions $f_{x1}(x,y)$ and $f_{y1}(x,y)$ defined so that they are null on the substructure boundary and constant over the circular domain. Thus the positions of the nodes of the substructures are moved as follows:

$$x_j^s = x_j^0 + \epsilon_{x1}^s f_{x1}(x_j^0,y_j^0) \quad , \quad y_j^s = y_j^0 + \epsilon_{y1}^s f_{y1}(x_j^0,y_j^0), \tag{40}$$

where ϵ_{x1}^s and ϵ_{y1}^s are mesh parameters which are here defined as independent random variables with dispersion parameters δ_{x1} and δ_{y1} where $\delta_{x1} = \delta_{y1}$. The values of these dispersion parameters are not given here. Indeed what matters in Eq. (40) is the product between the dispersion parameters and the

shape functions, i.e., the only values of δ_{x1} and δ_{y1} could be misunderstood. A schematic of the distorted substructure meshes obtained by considering these dispersion parameters is given in Fig. 8.

In addition to mesh distortion, it is assumed that the thickness of the circular part of each substructure is subjected to changes, i.e.,

$$h_2^s = h_2^0 + \epsilon_h^s, \tag{41}$$

with $h_2^0 = 0.005$ m and ϵ_h^s a random variable with support $[-\delta_h\,,\,\delta_h]$ with $\delta_h = 0.1h_2^0$. Hence, each substructure model depends on three parameters ϵ_{x1}^s, ϵ_{y1}^s and ϵ_h^s as opposed to the previous example where only two parameters were of concern. To address this problem, a 3D interpolation scheme based on $3 \times 3 \times 3$ interpolation points and 27 quadratic Lagrange polynomials $N_p(\xi^s, \eta^s, \zeta^s)$ $(p = 1, \ldots, 27)$ with $\xi^s = \sqrt{5/3}(\epsilon_{x1}^s/\delta_{x1})$, $\eta^s = \sqrt{5/3}(\epsilon_{y1}^s/\delta_{y1})$ and $\zeta^s = \sqrt{5/3}(\epsilon_h^s/\delta_h)$ are considered. For the sake of clarity, a schematic of the distorted meshes of the substructures at interpolation points in the (ξ^s, η^s) space is given in Fig. 9.

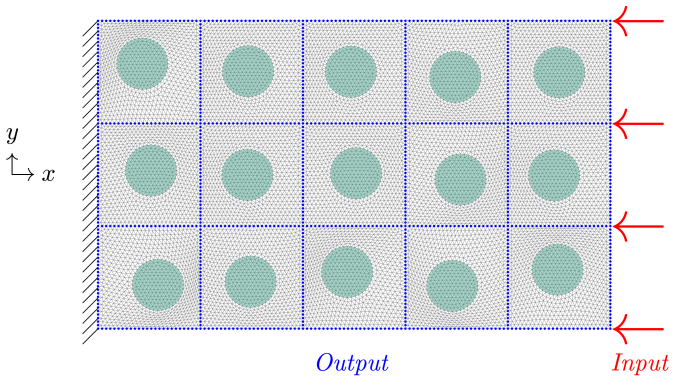


Figure 8: Nearly periodic structure with 2D substructures consisting of circular cores (in green) with varying positions and thicknesses. Blue dots indicate interface nodes.

Table 3: Material properties of the substructures (second example).

	Density (kg.m^{-3})	Young's modulus (GPa)	Poisson's ratio
First material (gray):	2700	70	0.35
Second material (green):	19, 250	340	0.27

The harmonic response of the structure is assessed over $[0\,,\,10,000]$ Hz. The structure is clamped at its left boundary and is excited at its right boundary by four unit forces acting leftwards at the substructure corners. In the present case, the quadratic velocity of the interface nodes (see blue dots in Fig. 8) — i.e., $\omega^2\|\mathbf{u}_{\mathtt{out}}\|^2$ with $\mathbf{u}_{\mathtt{out}}$ the displacement vector of the interface nodes (including $x-$ and $y-$displacement components) — is analyzed. FE results for the nearly periodic structure and the equivalent purely periodic one are shown in Fig. 10. Here, the damping coefficients of the substructures are $a = 0.01$ s^{-1} and $b = 10^{-8}$ s, see Eq. (4). In this example again, the two solutions appears to be dissimilar. In the frequency range analyzed, the substructures exhibit resonant behavior. Especially the first three eigenfrequencies of the baseline substructure with clamped boundaries are $f_1 \approx f_2 = 6033$ Hz and $f_3 = 8512$ Hz which correspond to band gap phenomena for the purely periodic structure. In contrast, in the nearly periodic case, the substructures have different eigenfrequencies which, as such, yield several local resonance peaks and a more irregular response function.

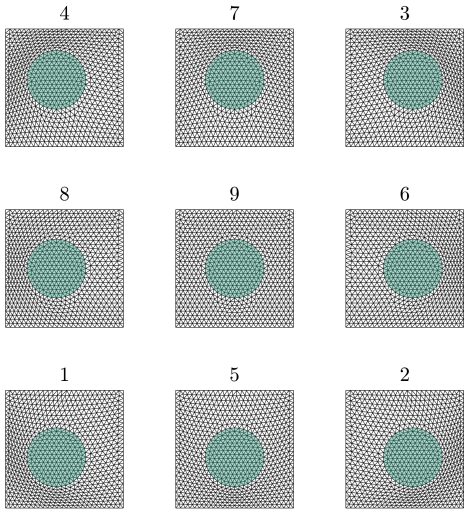


Figure 9: FE meshes of the substructures at interpolation points (second example).

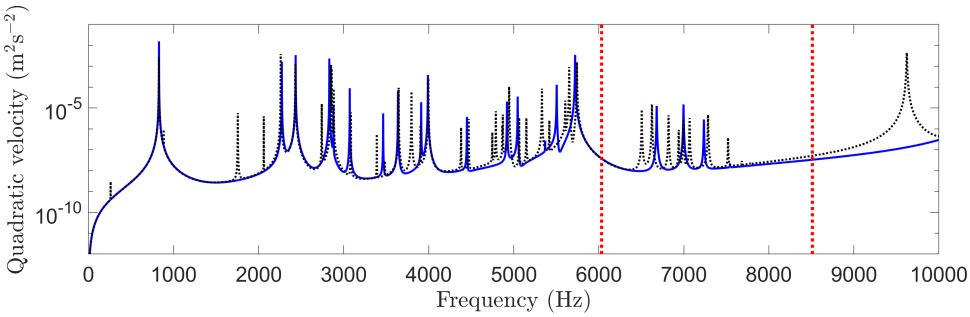


Figure 10: Frequency responses (second example): reference solutions for the purely periodic structure (blue solid line) and the nearly periodic structure (black dotted line). Red dotted vertical lines indicate the eigenfrequencies of the baseline substructure with clamped boundary.

Consider now the classic interpolation strategy, see Sec. 2. In this case, $M_{\mathrm{I}} = 3$ fixed interface modes of the substructures and $200 + 4$ interface modes of the structure — which represent the classic interface modes and four additional residual terms corresponding to the excitation DOFs, see Eq. (23) — are considered. The procedure to select the number of fixed interface modes of the substructures has been explained in Sec. 4.1. As for the choice of the number of interface modes of the structure, it can be determined through a sensitivity analysis of the harmonic response of the purely periodic structure. The solution issued from the interpolation strategy is compared with the reference solution in Fig. 11. As expected, dissimilarities between these solutions occur at high frequencies, i.e., after 5000 Hz.

On the other hand, the interpolation strategy with basis enrichment can be considered, see Secs. 3.1 and 3.2. Here, $M^{\mathsf{e}}_{\mathsf{st}} = 15$ high-order static modes are used to enrich the CB basis, see Eq. (33). Also, 204 static correction vectors are used to enrich the basis of interface modes (same number as interface modes, see Eq. (38)). With the basis enrichment techniques, the dynamic response of the nearly periodic structure can be computed as shown in Fig. 12. In this case, the proposed solution perfectly matches the reference one, even at high frequencies. This therefore demonstrates the relevance of the proposed strategy. In this case again, the interpolation strategies (without or with basis enrichment) yield more than 90% computational time saving. For the sake of clarity, the sizes of the full FE model and the reduced models are detailed in Tab. 4.

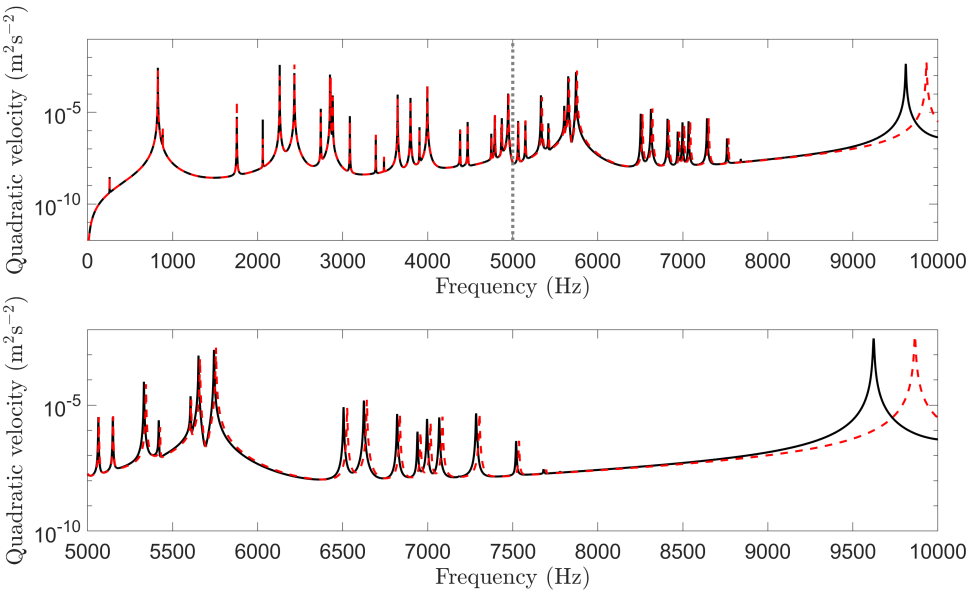


Figure 11: Frequency responses of the nearly periodic structure (second example): reference solution (black solid line) and proposed solution without basis enrichment (red dashed line) over $[0\,,\,10,000]$ Hz (top) and $[5000\,,\,10,000]$ Hz (bottom).

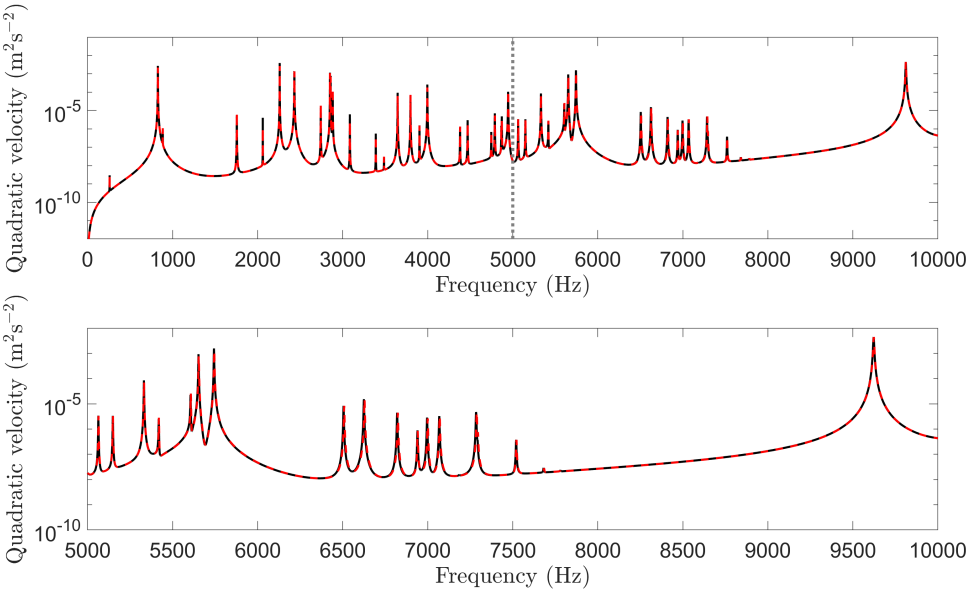


Figure 12: Frequency responses of the nearly periodic structure (second example): reference solution (black solid line) and proposed solution with basis enrichment (red dashed line) over $[0\,,\,10,000]$ Hz (top) and $[5000\,,\,10,000]$ Hz (bottom).

4.3. 3D substructures

As a last example, a nearly periodic structure with 4×4 3D substructures is considered as shown in Fig. 13. Each substructure consists of a square-base of surface area 0.07×0.07 m^2 and thickness 0.002 m supporting a hanger arm — i.e., three straight parts of same thickness $L^e = 0.7/21$ m ($x-$direction) with respective lengths $11L^e$, $3L^e$ and $7L^e$ and respective widths L^e, L^e and $2L^e$, see Fig. 1 — which behaves as a multi-DOF resonator. The related material properties (only one material) are listed in Tab. 5. In the present case, the FE mesh of a baseline substructure (no distortion) is generated with COMSOL. Afterwards the FE meshes and FE models of the distorted substructures are generated with MATLAB

Table 4: Number of DOFs involved in the models (second example).

	Reference	Interpolated	Interpolated with basis enrichment
Substructure	Boundary: 216 Internal: 1452	Static modes: 216 Fixed interface modes: 3	Static modes: 216 Fixed interface modes: 3 High-order static modes: 15
Interface	1840	Interface modes: 204	Interface modes: 204 Static correction vectors: 204
Structure	$23,620$	$204 + 15 \times 3 = 249$	$408 + 15 \times (3 + 15) = 678$

using in-house codes. Here, the substructure FE mesh is based on 10-node quadratic tetrahedrons and involves $16,533$ internal DOFs and 1512 boundary DOFs which is substantial in comparison with the previous examples. Mesh variations are introduced which involve changing the angle of the hanger as well as moving its position in the (x, y) plane. Thus three mesh parameters are introduced, namely ϵ_1^s for the angle, and ϵ_{x2}^s and ϵ_{y2}^s for the position along the $x-$ and $y-$ directions (respectively). These parameters are associated with four shape functions, i.e., $f_{x1}(x, y)$ and $f_{y1}(x, y)$ for ϵ_1^s, and $f_{x2}(x, y)$ and $f_{y2}(x, y)$ for ϵ_{x2}^s and ϵ_{y2}^s (respectively). In this case the nodes of the baseline substructure are moved as follows:

$$x_j^s = x_j^0 + \epsilon_1^s f_{x1}(x_j^0, y_j^0) + \epsilon_{x2}^s f_{x2}(x_j^0, y_j^0) \quad , \quad y_j^s = y_j^0 + \epsilon_1^s f_{y1}(x_j^0, y_j^0) + \epsilon_{y2}^s f_{y2}(x_j^0, y_j^0). \quad (42)$$

Again, a 3D interpolation scheme based on $3\times3\times3$ interpolation points and 27 quadratic Lagrange polynomials $N_p(\xi^s, \eta^s, \zeta^s)$ $(p = 1, \ldots, 27)$ with $\xi^s = \sqrt{5/3}(\epsilon_1^s/\delta_1)$, $\eta^s = \sqrt{5/3}(\epsilon_{x2}^s/\delta_{x2})$ and $\zeta^s = \sqrt{5/3}(\epsilon_{y2}^s/\delta_{y2})$ (δ_1, δ_{x2} and δ_{y2} being the dispersion parameters which are used to obtain, for instance, the distorted substructures shown in Fig. 13) is considered. For the sake of clarity, a schematic of the distorted meshes of the substructures at some interpolation points (9 points among 27) is given in Fig. 14.

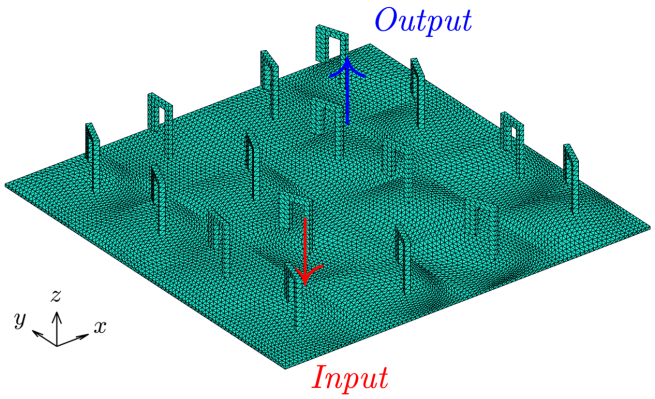


Figure 13: Nearly periodic structure with 3D substructures.

Table 5: Material properties of the substructures (third example).

	Density (kg.m^{-3})	Young's modulus (GPa)	Poisson's ratio
Material:	2700	70	0.3

The nearly periodic structure is assumed to be clamped at its four edges, and it is excited by a unit harmonic force acting downward (input) at some substructure corner as shown in Fig. 13. The resulting quadratic velocity (output) — i.e., $\omega^2|w_{\mathtt{out}}|^2$ with $w_{\mathtt{out}}$ the $z-$displacement — at another corner (see Fig. 13) is assessed. The damping coefficients of the substructures are similar to those considered in

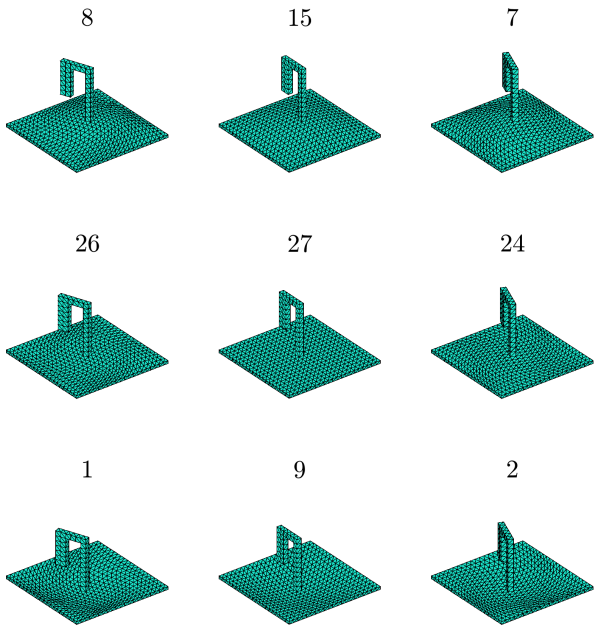


Figure 14: FE meshes of the substructures at interpolation points (third example).

Sec. 4.1, i.e., $a = 0.01$ s^{-1} and $b = 10^{-7}$ s. First, the FE-based reference frequency responses of the nearly and purely periodic structures are computed over $[0\,,\,2500]$ Hz as shown in Fig. 15. In this frequency range, the baseline substructure reveals four eigenfrequencies $f_1 = 722$ Hz, $f_2 = 760$ Hz, $f_3 = 1468$ Hz and $f_4 = 1573$ Hz which are associated with band gap phenomena for the purely periodic structure. Especially the band gap effect appears to be well pronounced around f_1 and f_2 where low vibration levels occur. Aside from band gaps, the frequency response of the purely periodic structure appears to be regular with well-defined resonance peaks. In contrast the quadratic velocity of the nearly periodic structure shows irregular behavior, e.g., after 1000 Hz where many local resonance peaks occur. These peaks are caused by the dissimilarities between the parameters of the substructures.

By considering the classic interpolation strategy, the harmonic behavior of the nearly periodic structure can be predicted as shown in Fig. 16. Here, $M_{\mathtt{I}} = 6$ fixed interface modes of the substructures are retained together with $100 + 1$ interface modes of the structure. Again, dissimilarities between the proposed and reference solutions can be observed at high frequencies, i.e., after 1000 Hz (see Fig. 16). In particular, it appears that the prediction of the last resonance frequencies of the substructure is prone to strong errors.

To improve the accuracy of the interpolation strategy, the basis enrichment techniques proposed in Secs. 3.1 and 3.2 are considered. Here, $M^{\mathtt{e}}_{\mathtt{st}} = 12$ high-order static modes for the CB basis (Eq. (33)) and 101 static correction vectors for the basis of interface modes (Eq. (38)) are used. The resulting predicted response function is shown in Fig. 17. Again, it is seen that the proposed solution is in good agreement with the reference one over the whole frequency band. Especially the resonance peaks of the structure can be well predicted. Slight errors can be observed close to the band gap regions, which are however associated with low vibration levels. In terms of computational time, the interpolation strategies yield a drastic reduction, i.e., around 99.9% time saving to compute the frequency response. In this example indeed, the numbers of substructure DOFs and interface DOFs can be strongly reduced, see Tab. 6. As such the proposed approach could be advantageously used to perform parametric analysis like in optimization procedure, e.g., to seek optimal design for tunable metamaterials.

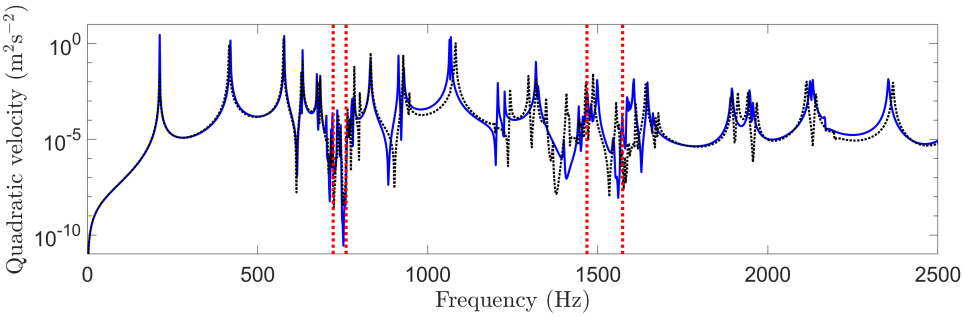


Figure 15: Frequency responses (third example): reference solutions for the purely periodic structure (blue solid line) and the nearly periodic structure (black dotted line). Red dotted vertical lines indicate the eigenfrequencies of the baseline substructure with clamped boundary.

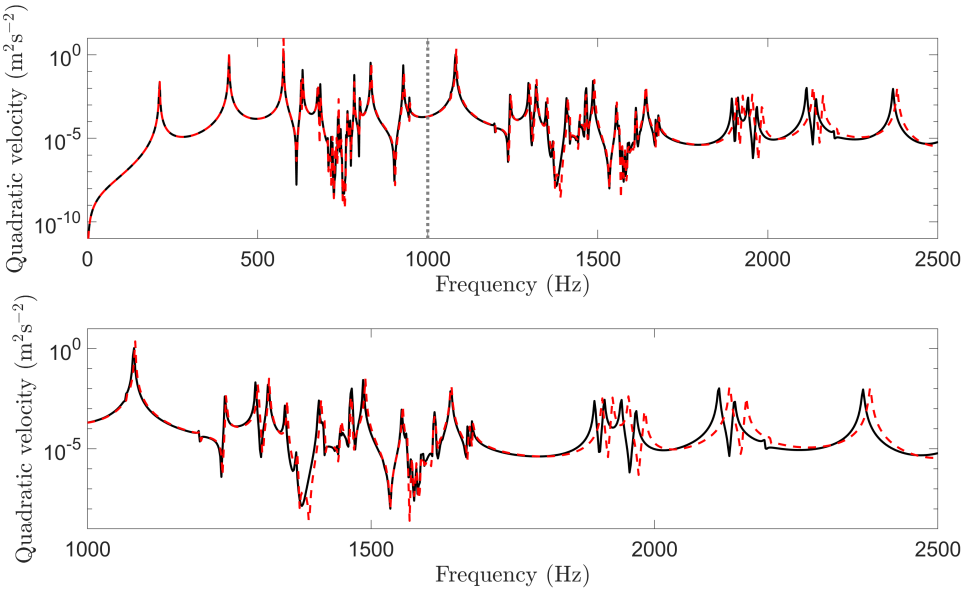


Figure 16: Frequency responses of the nearly periodic structure (third example): reference solution (black solid line) and proposed solution without basis enrichment (red dashed line) over $[0\,,\,2500]$ Hz (top) and $[1000\,,\,2500]$ Hz (bottom).

Table 6: Number of DOFs involved in the models (third example).

	Reference	Interpolated without enrichment	Interpolated with enrichment
Substructure	Boundary: 1512 Internal: $16,533$	Static modes: 1512 Fixed interface modes: 6	Static modes: 1512 Fixed interface modes: 6 High-order static modes: 12
Interface	8937	Interface modes: 101	Interface modes: 101 Static correction vectors: 101
Structure	$273,465$	$101 + 16 \times 6 = 197$	$202 + 16 \times (6 + 12) = 490$

5. Conclusion

The dynamic behavior of nearly periodic structures has been analyzed via parametric model reduction. The focus is on the study of complex 2D or 3D substructures subjected to geometric modifications including mesh variations or more classic parametric changes. To model a nearly periodic structure with parameter-dependent substructures, an interpolation strategy can be considered which works two ways: (i) interpolation of the CB reduced substructure matrices over a parametric space; (ii) interface reduction,

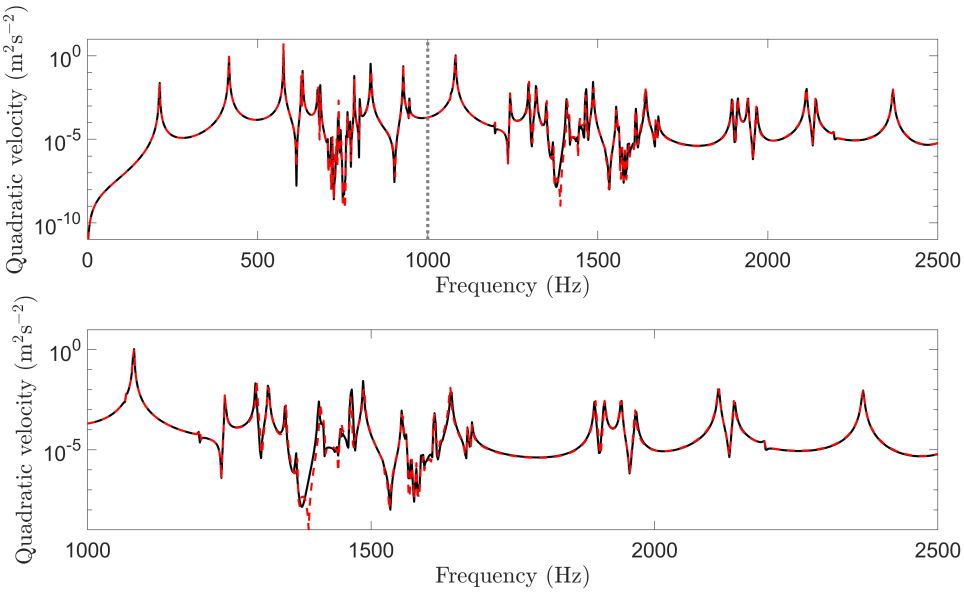


Figure 17: Frequency responses of the nearly periodic structure (third example): reference solution (black solid line) and proposed solution with basis enrichment (red dashed line) over $[0\,,\,2500]$ Hz (top) and $[1000\,,\,2500]$ Hz (bottom).

between the substructures, using the interface modes of the equivalent purely periodic structure. In this paper, two basis enrichment techniques have been proposed to improve the accuracy of the interpolation strategy. This consists in (i) using high-order static modes to enrich the CB basis of the substructures and (ii) using static correction vectors which account for parametric changes to enrich the basis of interface modes. Numerical experiments have been carried out which have clearly demonstrated the accuracy and efficiency (CPU times) of the proposed approach for predicting the harmonic behavior of nearly periodic structures with complex substructures.

A. Appendix A. Proof of Eq. (11) (correlation criterion)

Let us consider the following minimization problem:

$$\min_{\widetilde{\mathbf{Q}}_p} \|(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\widetilde{\mathbf{X}}_p - (\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\widetilde{\mathbf{X}}^0\,\widetilde{\mathbf{Q}}_p\|_F^2, \tag{A-1}$$

for some full rank matrix $\widetilde{\mathbf{Q}}_p$, where matrix $(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\widetilde{\mathbf{X}}^0$ is orthogonal by definition [5]. In Eq. (A-1), $\widetilde{\mathbf{Q}}_p$ performs a change of basis to make the column vectors of $(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\widetilde{\mathbf{X}}^0$ and $(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\widetilde{\mathbf{X}}_p$ "as close as possible". Eq. (A-1) represents a least squares problem for the Frobenius norm whose solution is:

$$\widetilde{\mathbf{Q}}_p = [(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\widetilde{\mathbf{X}}^0]^+(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\widetilde{\mathbf{X}}_p, \tag{A-2}$$

where $[(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\widetilde{\mathbf{X}}^0]^+$ is the left pseudo-inverse of $(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\widetilde{\mathbf{X}}^0$. Recall that the left pseudo-inverse of a full (column) rank matrix $\mathbf{A}$ is expressed as $\mathbf{A}^+ = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T$. In the present case, this yields $[(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\widetilde{\mathbf{X}}^0]^+ = [(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\widetilde{\mathbf{X}}^0]^T$ since $(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\widetilde{\mathbf{X}}^0$ has full column rank [6] and $(\widetilde{\mathbf{X}}^0)^T\mathbf{M}^0_{\mathtt{II}}\widetilde{\mathbf{X}}^0 = \mathbf{I}$ from

[5] Indeed $(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}$ is symmetric and $(\widetilde{\mathbf{X}}^0)^T\mathbf{M}^0_{\mathtt{II}}\widetilde{\mathbf{X}}^0 = \mathbf{I}$ from the properties of the modes.

[6] Indeed $(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}$ is positive definite since $\mathbf{M}^0_{\mathtt{II}}$ is symmetric and positive definite; also $\widetilde{\mathbf{X}}^0$ has full column rank.

the orthogonality and normalization properties of the fixed interface modes. Since $[(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\widetilde{\mathbf{X}}^0]^T = (\widetilde{\mathbf{X}}^0)^T(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}$ (with $(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}$ symmetric), one obtains:

$$\widetilde{\mathbf{Q}}_p = (\widetilde{\mathbf{X}}^0)^T\mathbf{M}^0_{\mathtt{II}}\widetilde{\mathbf{X}}_p = \mathbf{\Psi}^T\widetilde{\mathbf{X}}_p, \tag{A-3}$$

with $\mathbf{\Psi} = \mathbf{M}^0_{\mathtt{II}}\widetilde{\mathbf{X}}^0$. Thus the following matrix $(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\widetilde{\mathbf{X}}^0\widetilde{\mathbf{Q}}_p = (\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\widetilde{\mathbf{X}}^0(\mathbf{\Psi}^T\widetilde{\mathbf{X}}_p)$ with column vectors close to those of $(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}\widetilde{\mathbf{X}}_p$ can be considered. Since $(\mathbf{M}^0_{\mathtt{II}})^{\frac{1}{2}}$ is square positive definite, this means that the column vectors of $\widetilde{\mathbf{X}}^0(\mathbf{\Psi}^T\widetilde{\mathbf{X}}_p)$ and $\widetilde{\mathbf{X}}_p$ are close. Conversely, the following alternative matrix of fixed interface modes $\widehat{\mathbf{X}}_p = \widetilde{\mathbf{X}}_p(\mathbf{\Psi}^T\widetilde{\mathbf{X}}_p)^{-1}$ with column vectors close to those of $\widetilde{\mathbf{X}}^0$ can be considered (which remains true provided that $\mathbf{\Psi}^T\widetilde{\mathbf{X}}_p$ has full rank).

B. Appendix B. Proof of Eq. (29)

From Eq. (5), the following exact expression of the displacement vector $\mathbf{u}^s_{\mathtt{I}}$ for the internal nodes of a substructure s can be proposed:

$$\mathbf{u}^s_{\mathtt{I}} = \mathbf{X}_{\mathtt{st}}\mathbf{u}^s_{\mathtt{B}} + \widetilde{\mathbf{X}}\widetilde{\boldsymbol{\alpha}}^s + (\mathbf{u}^s_{\mathtt{I}})_{\mathtt{r}}, \tag{B-1}$$

where $(\mathbf{u}^s_{\mathtt{I}})_{\mathtt{r}}$ denotes the contribution of the residual modes — i.e., those not retained in the reduced matrix of fixed interface modes $\widetilde{\mathbf{X}}$ — expressed as $(\mathbf{u}^s_{\mathtt{I}})_{\mathtt{r}} = \widetilde{\mathbf{X}}_{\mathtt{r}}\widetilde{\boldsymbol{\alpha}}^s_{\mathtt{r}}$ with $\widetilde{\mathbf{X}}_{\mathtt{r}}$ the matrix of residual fixed interface modes and $\widetilde{\boldsymbol{\alpha}}^s_{\mathtt{r}}$ the related vector of generalized coordinates. Then let us consider the dynamic equation for the internal DOFs of the undamped substructure, see Eqs. (3) and (8):

$$\left(-\omega^2\mathbf{M}_{\mathtt{II}} + \mathbf{K}_{\mathtt{II}}\right)\mathbf{u}^s_{\mathtt{I}} = -\left(-\omega^2\mathbf{M}_{\mathtt{IB}} + \mathbf{K}_{\mathtt{IB}}\right)\mathbf{u}^s_{\mathtt{B}}. \tag{B-2}$$

Inserting Eq. (B-1) into Eq. (B-2) gives:

$$\left(-\omega^2\mathbf{M}_{\mathtt{II}} + \mathbf{K}_{\mathtt{II}}\right)\left(\widetilde{\mathbf{X}}\widetilde{\boldsymbol{\alpha}}^s + \widetilde{\mathbf{X}}_{\mathtt{r}}\widetilde{\boldsymbol{\alpha}}^s_{\mathtt{r}}\right) = -\left(-\omega^2\mathbf{M}_{\mathtt{IB}} + \mathbf{K}_{\mathtt{IB}} - \omega^2\mathbf{M}_{\mathtt{II}}\mathbf{X}_{\mathtt{st}} + \mathbf{K}_{\mathtt{II}}\mathbf{X}_{\mathtt{st}}\right)\mathbf{u}^s_{\mathtt{B}}. \tag{B-3}$$

Since $\mathbf{X}_{\mathtt{st}} = -\mathbf{K}^{-1}_{\mathtt{II}}\mathbf{K}_{\mathtt{IB}}$ by definition, Eq. (B-3) simplifies to:

$$\left(-\omega^2\mathbf{M}_{\mathtt{II}} + \mathbf{K}_{\mathtt{II}}\right)\left(\widetilde{\mathbf{X}}\widetilde{\boldsymbol{\alpha}}^s + \widetilde{\mathbf{X}}_{\mathtt{r}}\widetilde{\boldsymbol{\alpha}}^s_{\mathtt{r}}\right) = \omega^2\left(\mathbf{M}_{\mathtt{IB}} + \mathbf{M}_{\mathtt{II}}\mathbf{X}_{\mathtt{st}}\right)\mathbf{u}^s_{\mathtt{B}}. \tag{B-4}$$

Let the modes $\widetilde{\mathbf{X}}_{\mathtt{r}}$ be normalized in such a way that $\widetilde{\mathbf{X}}^T_{\mathtt{r}}\mathbf{M}_{\mathtt{II}}\widetilde{\mathbf{X}}_{\mathtt{r}} = \mathbf{I}$ and $\widetilde{\mathbf{X}}^T_{\mathtt{r}}\mathbf{K}_{\mathtt{II}}\widetilde{\mathbf{X}}_{\mathtt{r}} = \widetilde{\mathbf{\Lambda}}_{\mathtt{r}}$ with $\widetilde{\mathbf{\Lambda}}_{\mathtt{r}}$ the diagonal matrix of the eigenvalues associated with these modes. Invoking mode orthogonality properties — i.e., $\widetilde{\mathbf{X}}^T_{\mathtt{r}}\mathbf{M}_{\mathtt{II}}\widetilde{\mathbf{X}} = \widetilde{\mathbf{X}}^T_{\mathtt{r}}\mathbf{K}_{\mathtt{II}}\widetilde{\mathbf{X}} = \mathbf{0}$ — and left-multiplying Eq. (B-4) by $\widetilde{\mathbf{X}}^T_{\mathtt{r}}$ yields:

$$\left(-\omega^2\mathbf{I} + \widetilde{\mathbf{\Lambda}}_{\mathtt{r}}\right)\widetilde{\boldsymbol{\alpha}}^s_{\mathtt{r}} = \omega^2\widetilde{\mathbf{X}}^T_{\mathtt{r}}\left(\mathbf{M}_{\mathtt{IB}} + \mathbf{M}_{\mathtt{II}}\mathbf{X}_{\mathtt{st}}\right)\mathbf{u}^s_{\mathtt{B}}. \tag{B-5}$$

Neglecting the dynamic term $-\omega^2\mathbf{I}$ in the left hand side of Eq. (B-5) yields:

$$\widetilde{\boldsymbol{\alpha}}^s_{\mathtt{r}} \approx \omega^2\widetilde{\mathbf{\Lambda}}^{-1}_{\mathtt{r}}\widetilde{\mathbf{X}}^T_{\mathtt{r}}\left(\mathbf{M}_{\mathtt{IB}} + \mathbf{M}_{\mathtt{II}}\mathbf{X}_{\mathtt{st}}\right)\mathbf{u}^s_{\mathtt{B}}, \tag{B-6}$$

and therefore:

$$(\mathbf{u}^s_{\mathtt{I}})_{\mathtt{r}} \approx \omega^2\widetilde{\mathbf{X}}_{\mathtt{r}}\widetilde{\mathbf{\Lambda}}^{-1}_{\mathtt{r}}\widetilde{\mathbf{X}}^T_{\mathtt{r}}\left(\mathbf{M}_{\mathtt{IB}} + \mathbf{M}_{\mathtt{II}}\mathbf{X}_{\mathtt{st}}\right)\mathbf{u}^s_{\mathtt{B}}. \tag{B-7}$$

Besides the spectral decomposition of the inverse of the matrix $\mathbf{K}_{\mathtt{II}}$ is expressed as $\mathbf{K}^{-1}_{\mathtt{II}} = \widetilde{\mathbf{X}}\widetilde{\mathbf{\Lambda}}^{-1}\widetilde{\mathbf{X}}^T + \widetilde{\mathbf{X}}_{\mathtt{r}}\widetilde{\mathbf{\Lambda}}^{-1}_{\mathtt{r}}\widetilde{\mathbf{X}}^T_{\mathtt{r}}$ with $\widetilde{\mathbf{\Lambda}}$ the diagonal matrix of eigenvalues for the retained modes. As a result, Eq. (B-7) leads to:

$$(\mathbf{u}^s_{\mathtt{I}})_{\mathtt{r}} \approx \omega^2\left(\mathbf{K}^{-1}_{\mathtt{II}} - \widetilde{\mathbf{X}}\widetilde{\mathbf{\Lambda}}^{-1}\widetilde{\mathbf{X}}^T\right)\left(\mathbf{M}_{\mathtt{IB}} + \mathbf{M}_{\mathtt{II}}\mathbf{X}_{\mathtt{st}}\right)\mathbf{u}^s_{\mathtt{B}} = \omega^2\mathbf{X}^{\mathtt{cor}}_{\mathtt{st}}\mathbf{u}^s_{\mathtt{B}}. \tag{B-8}$$

Eq. (B-8) means that the solution $(\mathbf{u}^s_{\mathtt{I}})_{\mathtt{r}}$ is located in the column space of $\mathbf{X}^{\mathtt{cor}}_{\mathtt{st}}$, as expected.

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